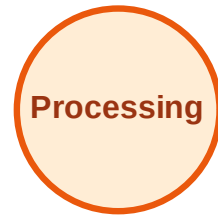


BFS Traversal

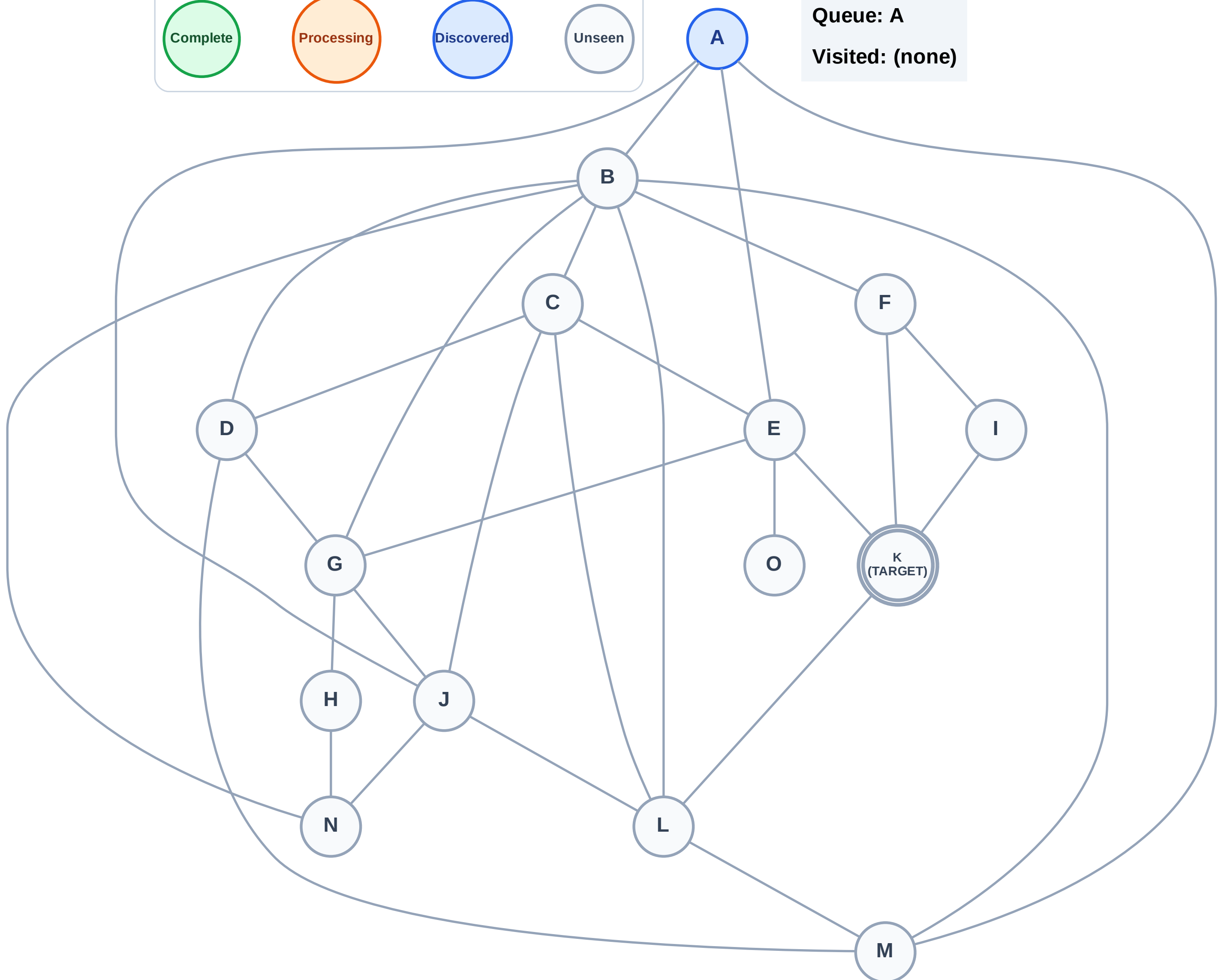
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

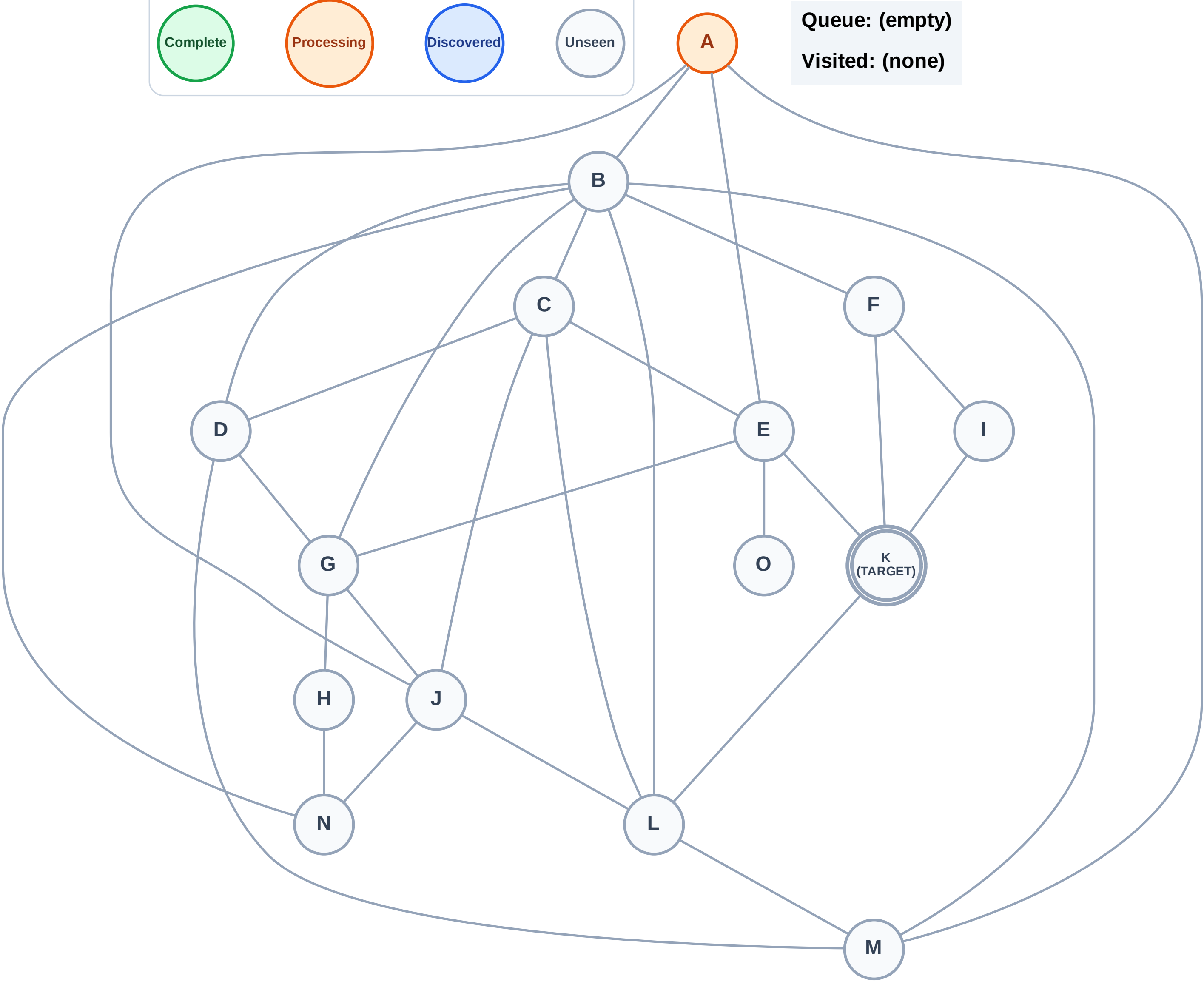
Complete

Processing

Discovered

Unseen

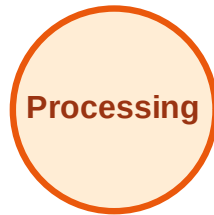
Queue: (empty)
Visited: (none)



BFS Traversal

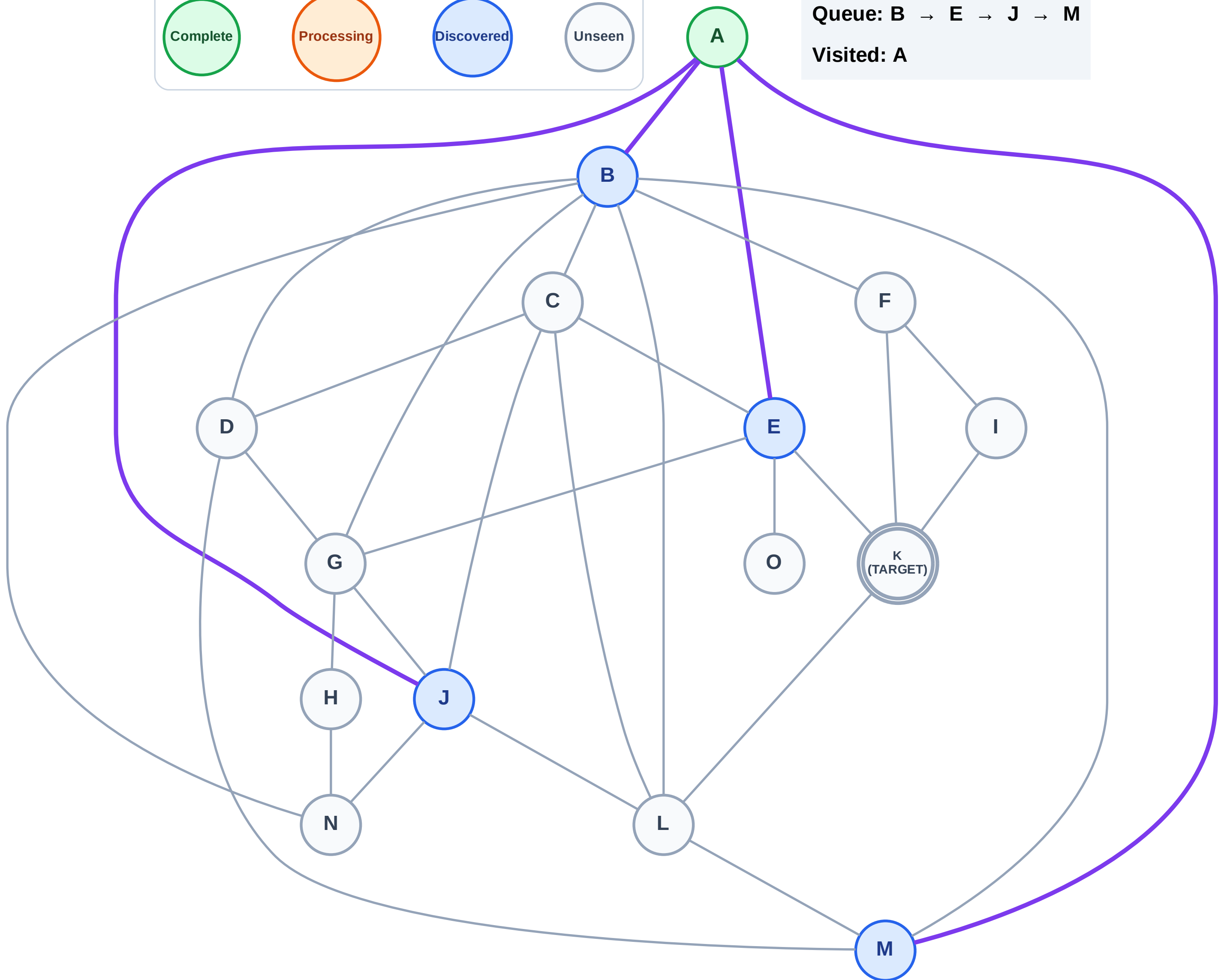
Step 3: Discover B, E, J, M from A

Legend



Queue: B → E → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

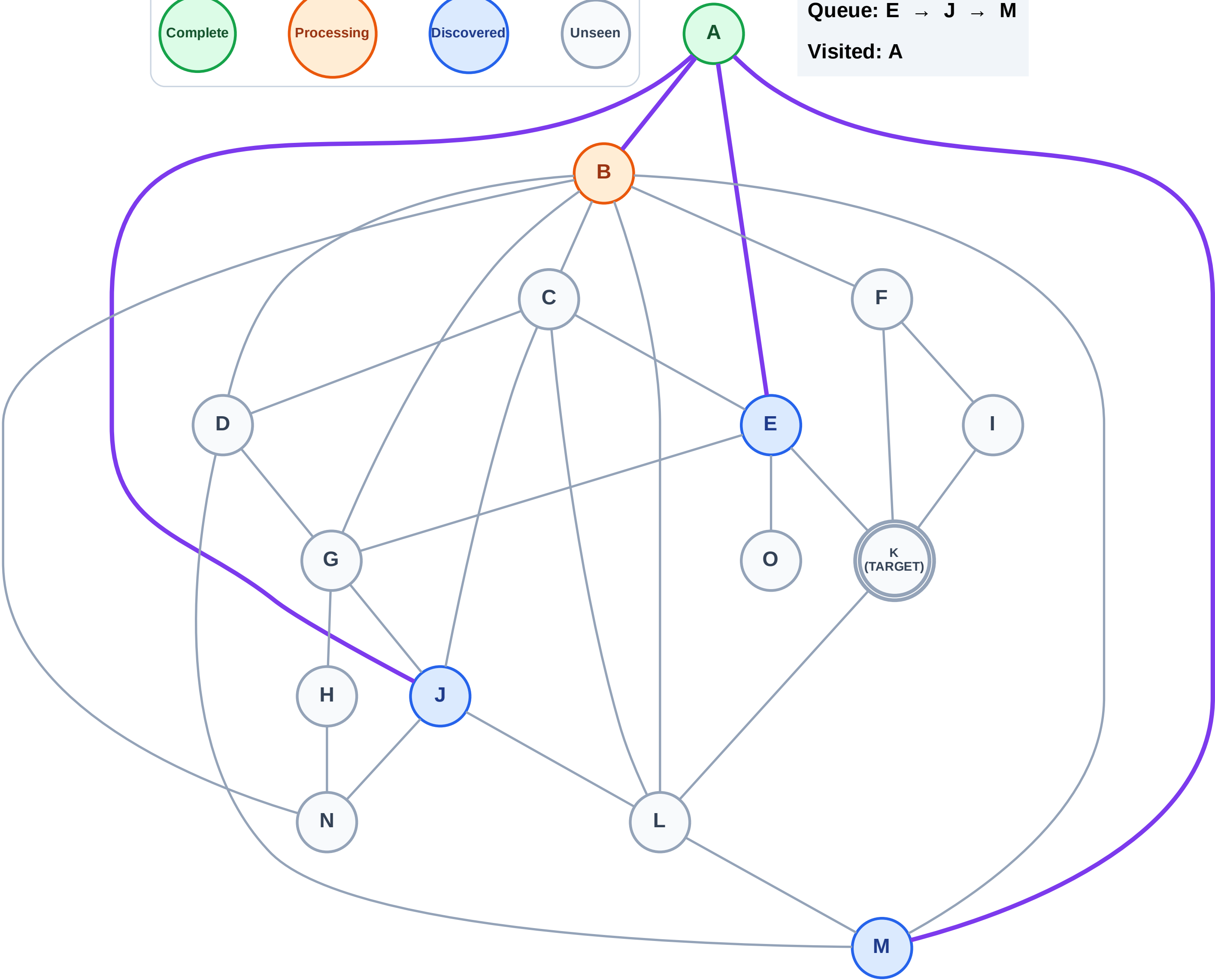
Processing

Discovered

Unseen

Queue: E → J → M

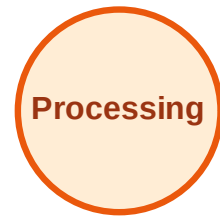
Visited: A



BFS Traversal

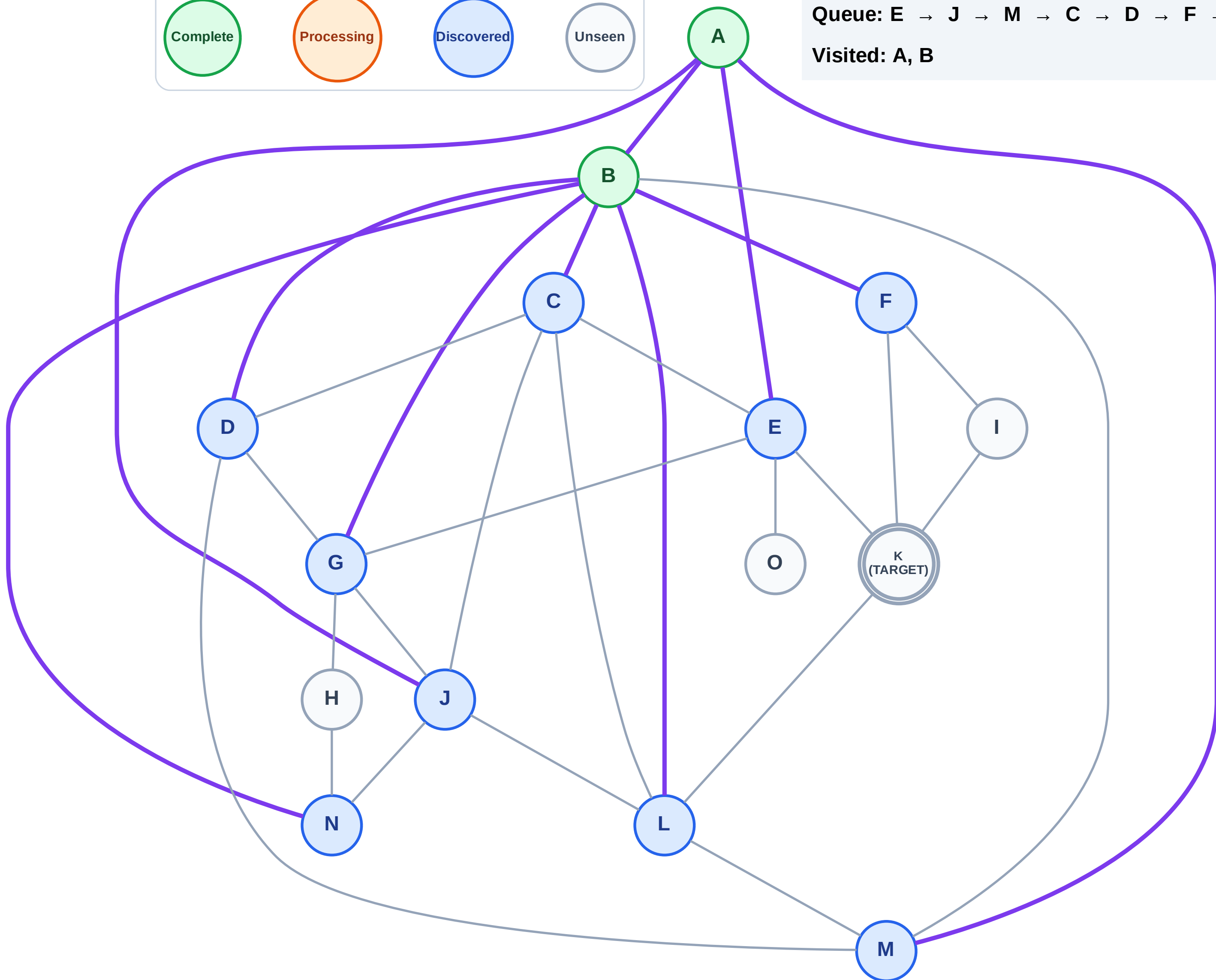
Step 5: Discover C, D, F, G, L, N from B

Legend



Queue: E → J → M → C → D → F → G → L → N

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

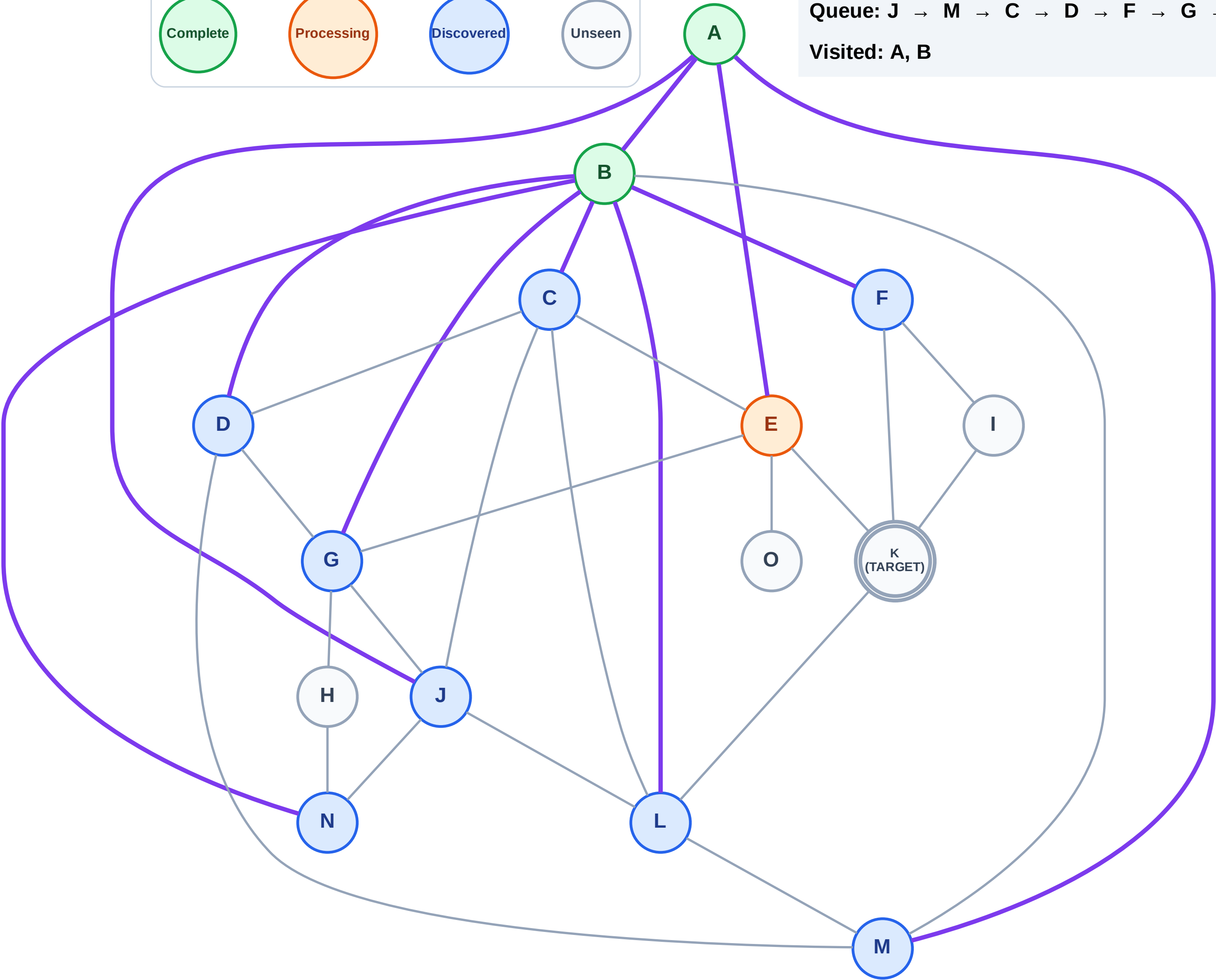
Processing

Discovered

Unseen

Queue: J → M → C → D → F → G → L → N

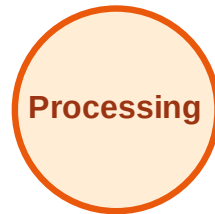
Visited: A, B



BFS Traversal

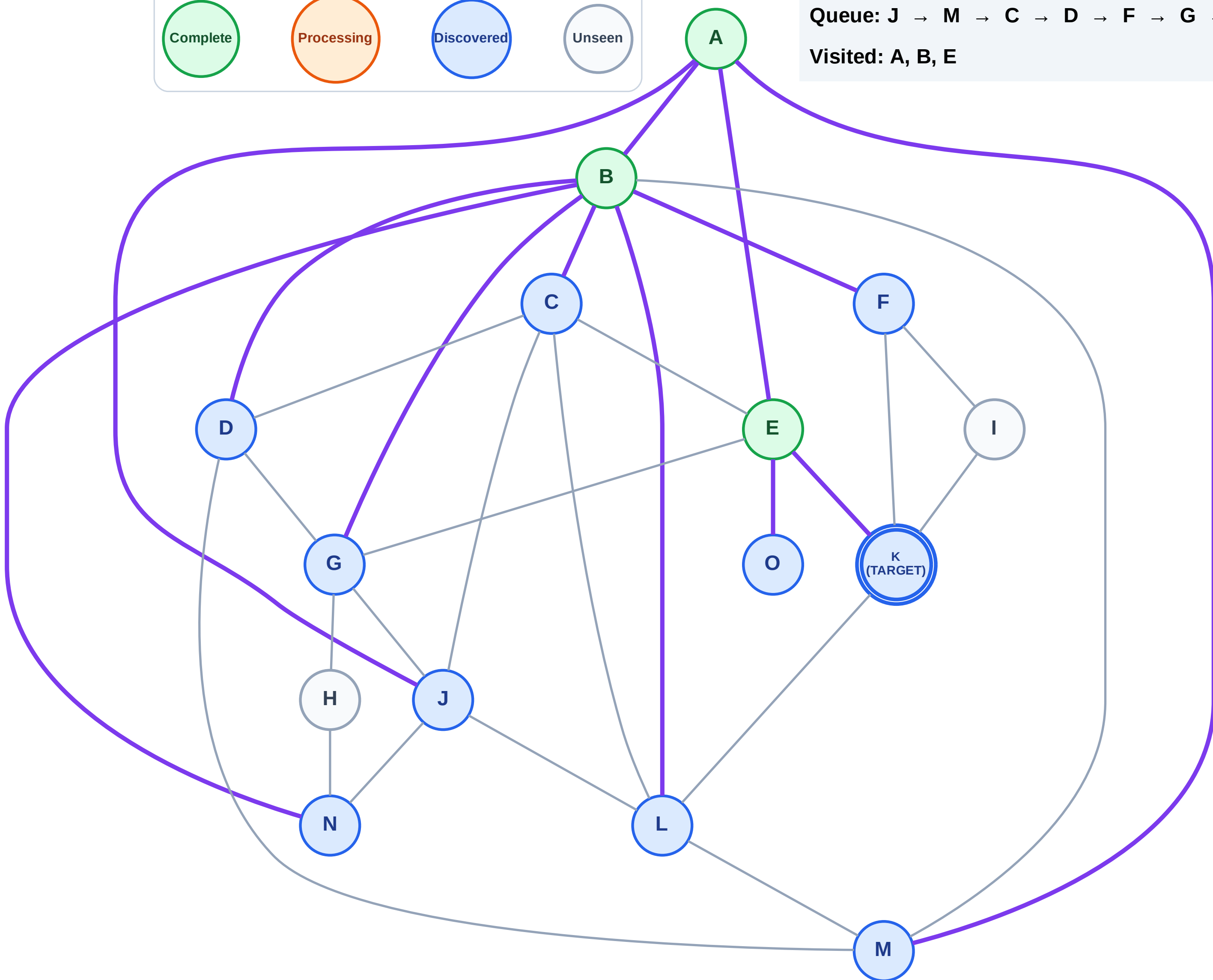
Step 7: Discover K, O from E

Legend



Queue: J → M → C → D → F → G → L → N → K → O

Visited: A, B, E



BFS Traversal

Step 8: Process node J

Legend

Complete

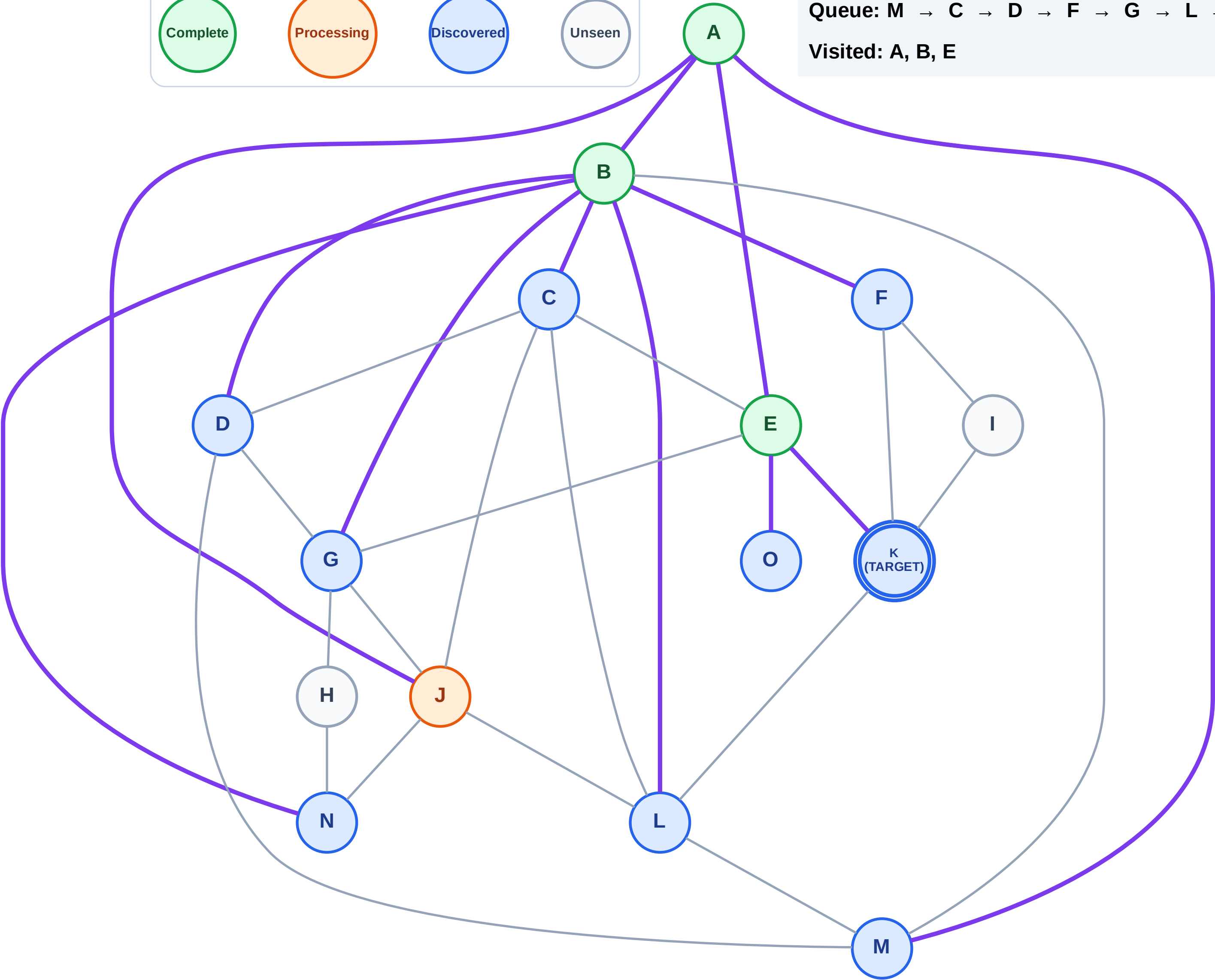
Processing

Discovered

Unseen

Queue: M → C → D → F → G → L → N → K → O

Visited: A, B, E



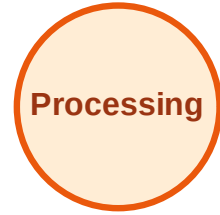
BFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing



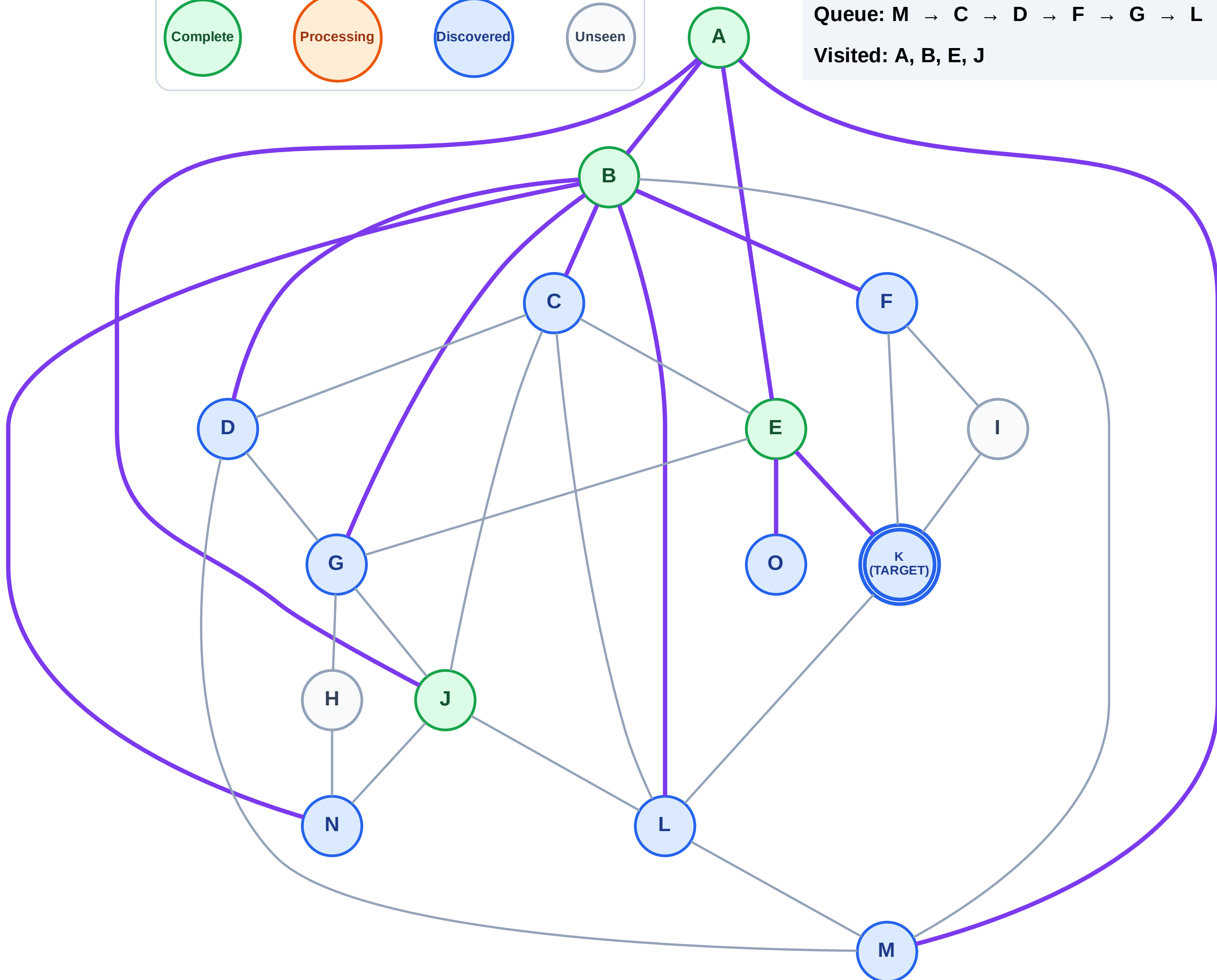
Discovered



Unseen

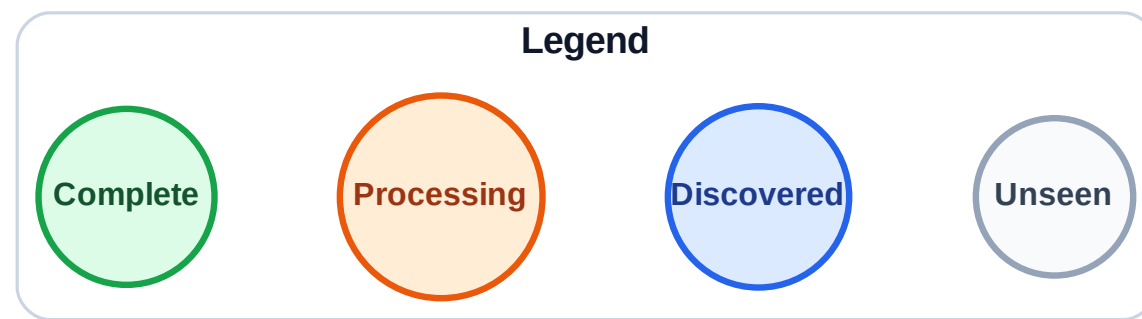
Queue: M → C → D → F → G → L → N → K → O

Visited: A, B, E, J



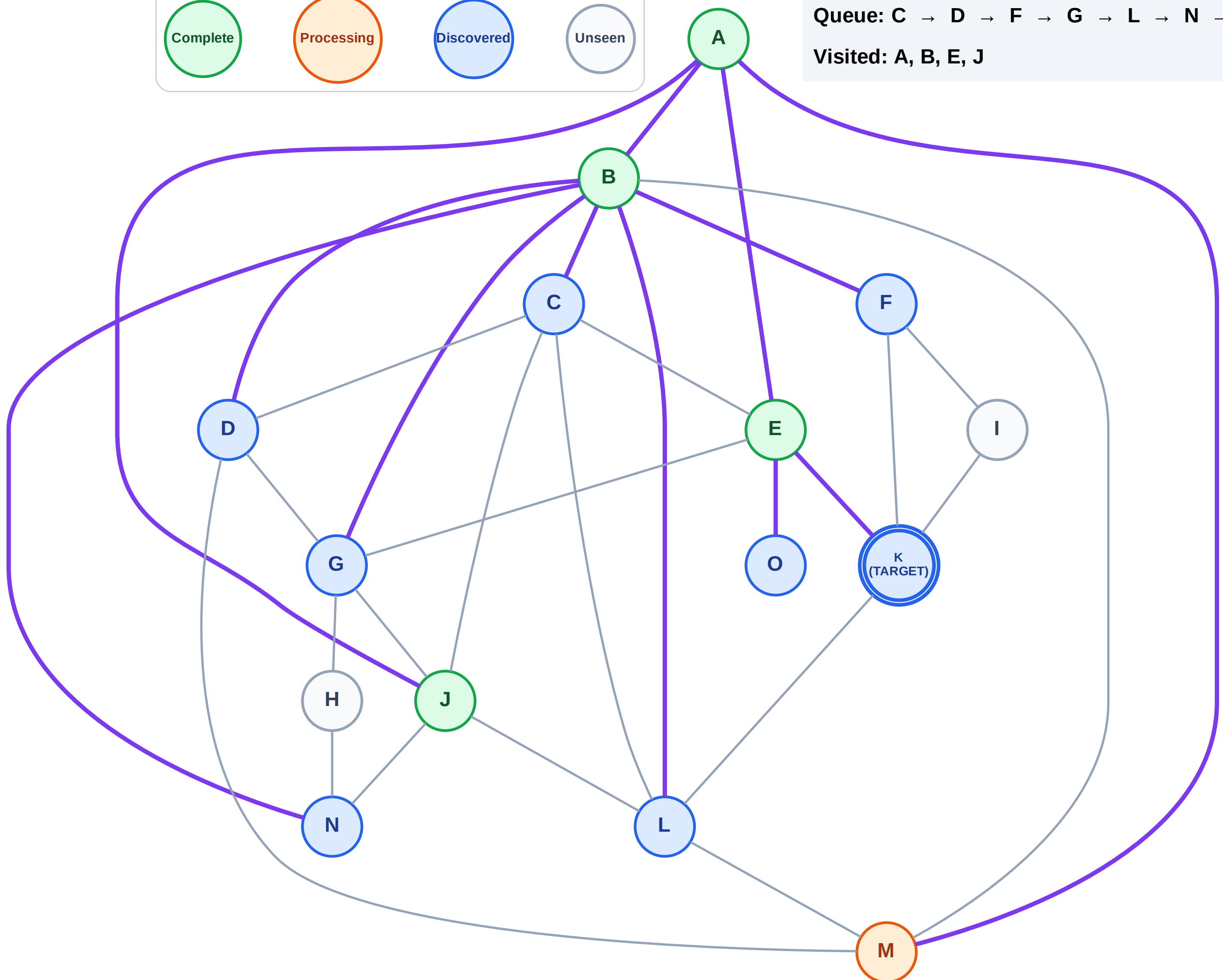
BFS Traversal

Step 10: Process node M



Queue: C → D → F → G → L → N → K → O

Visited: A, B, E, J



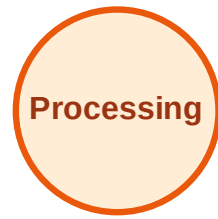
BFS Traversal

Step 11: No new neighbors from M

Legend



Complete



Processing



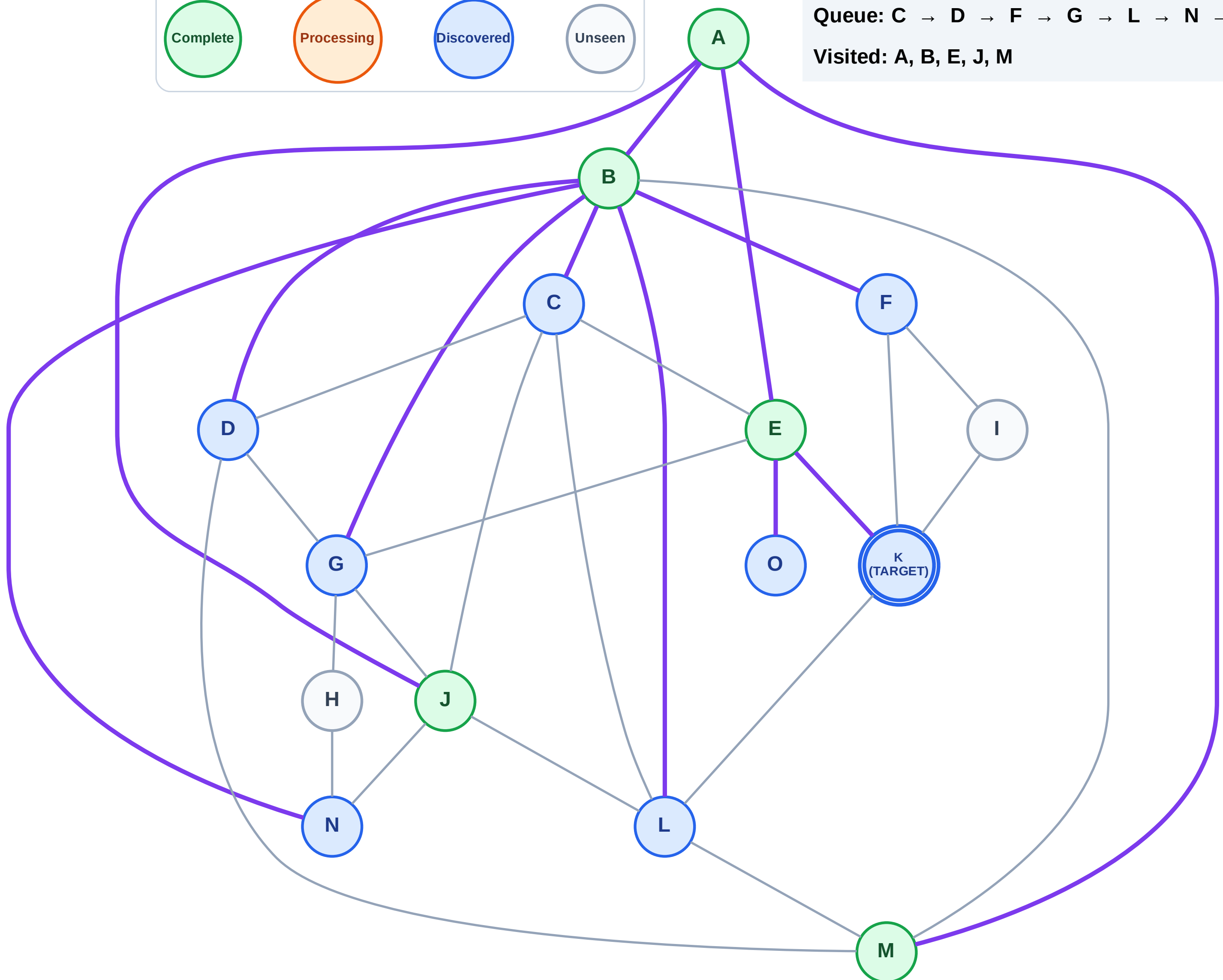
Discovered



Unseen

Queue: C → D → F → G → L → N → K → O

Visited: A, B, E, J, M



BFS Traversal

Step 12: Process node C

Legend

Complete

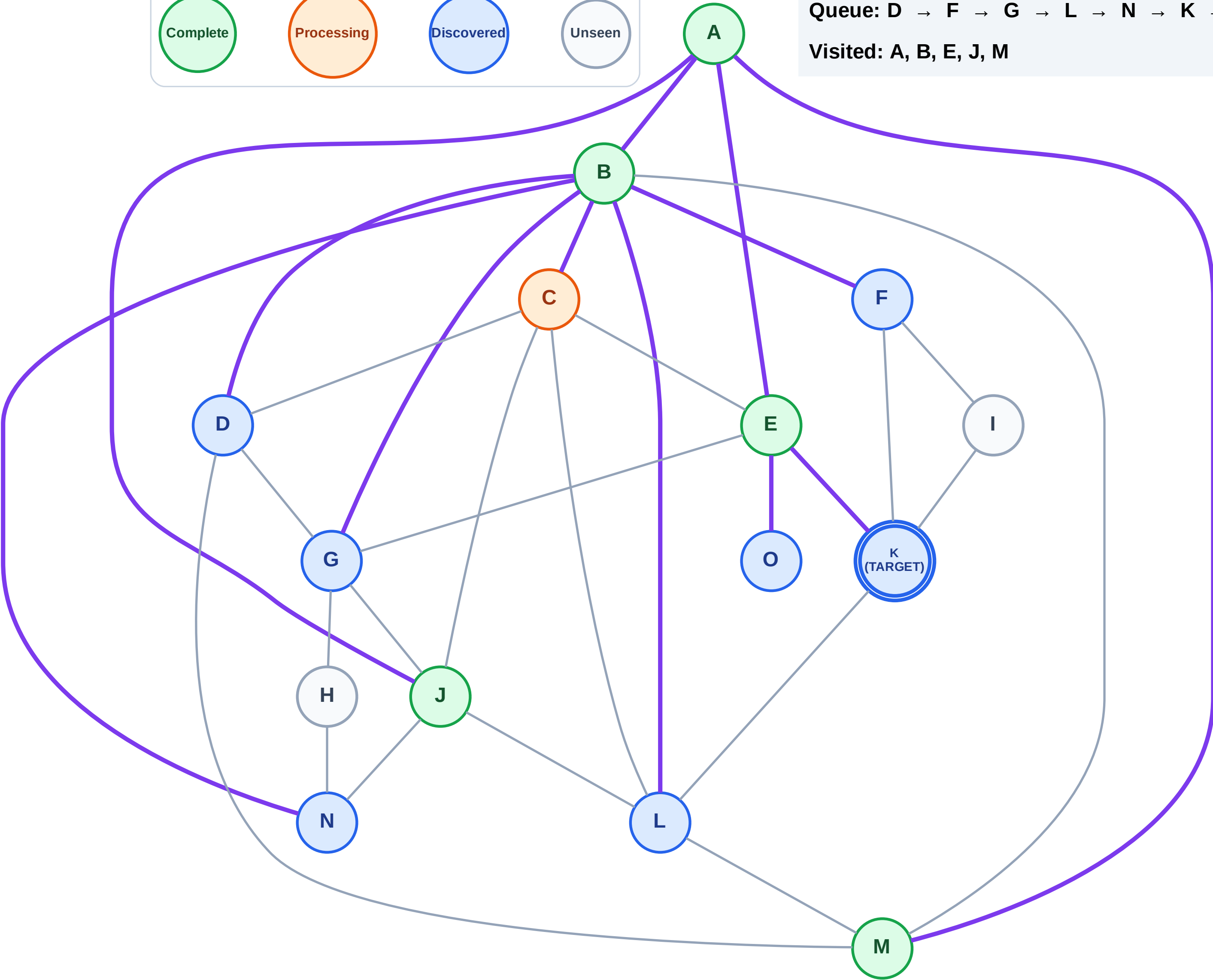
Processing

Discovered

Unseen

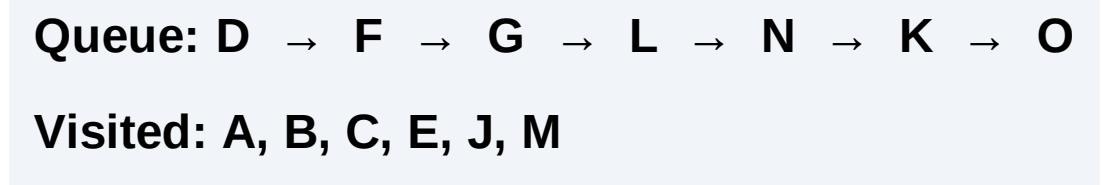
Queue: D → F → G → L → N → K → O

Visited: A, B, E, J, M



Step 13: No new neighbors from C

Step 13: No new neighbors from C



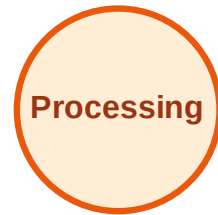
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



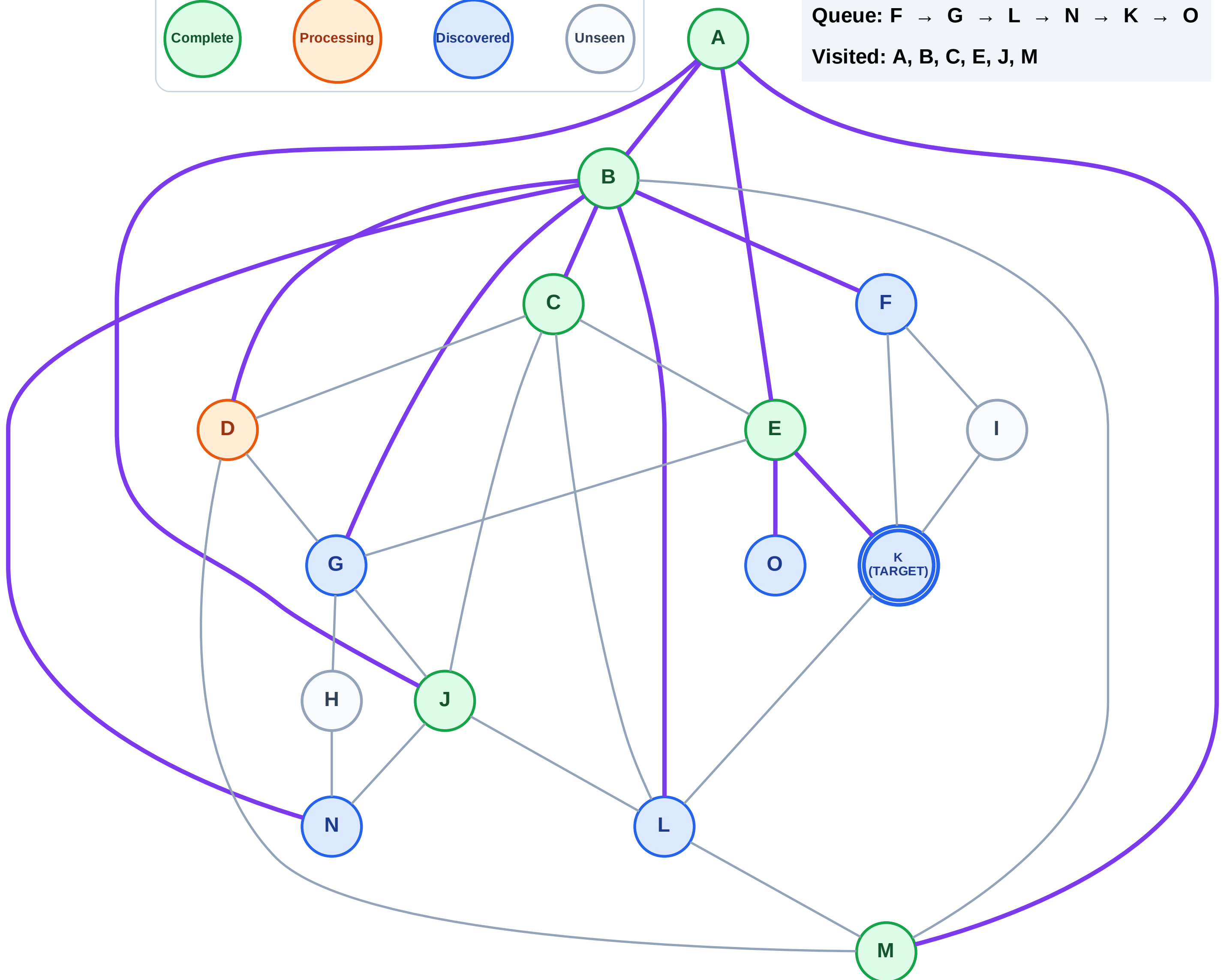
Discovered



Unseen

Queue: F → G → L → N → K → O

Visited: A, B, C, E, J, M



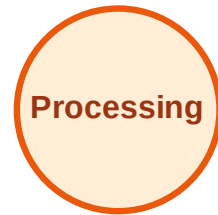
BFS Traversal

Step 15: No new neighbors from D

Legend



Complete



Processing



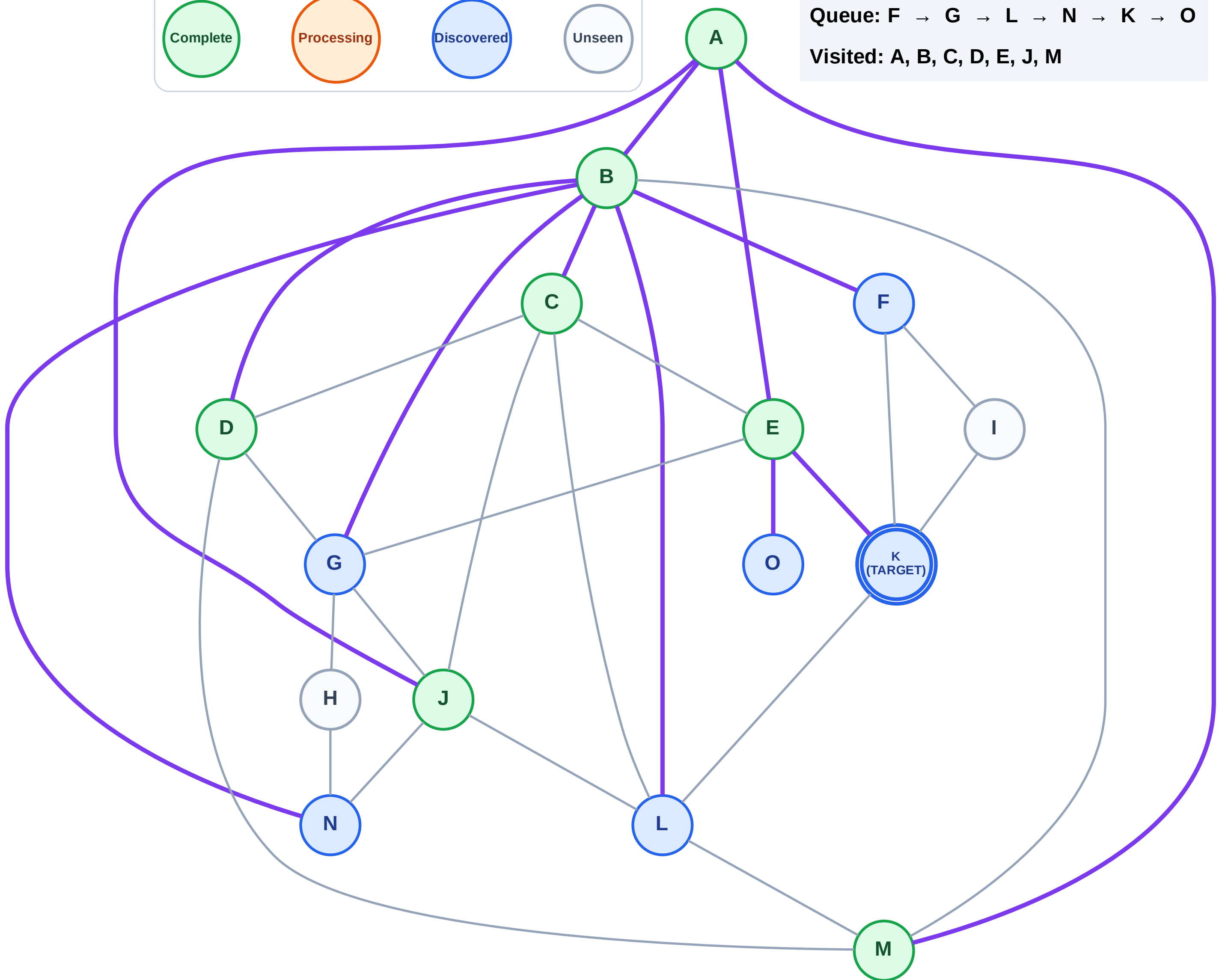
Discovered



Unseen

Queue: F → G → L → N → K → O

Visited: A, B, C, D, E, J, M



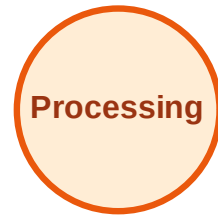
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



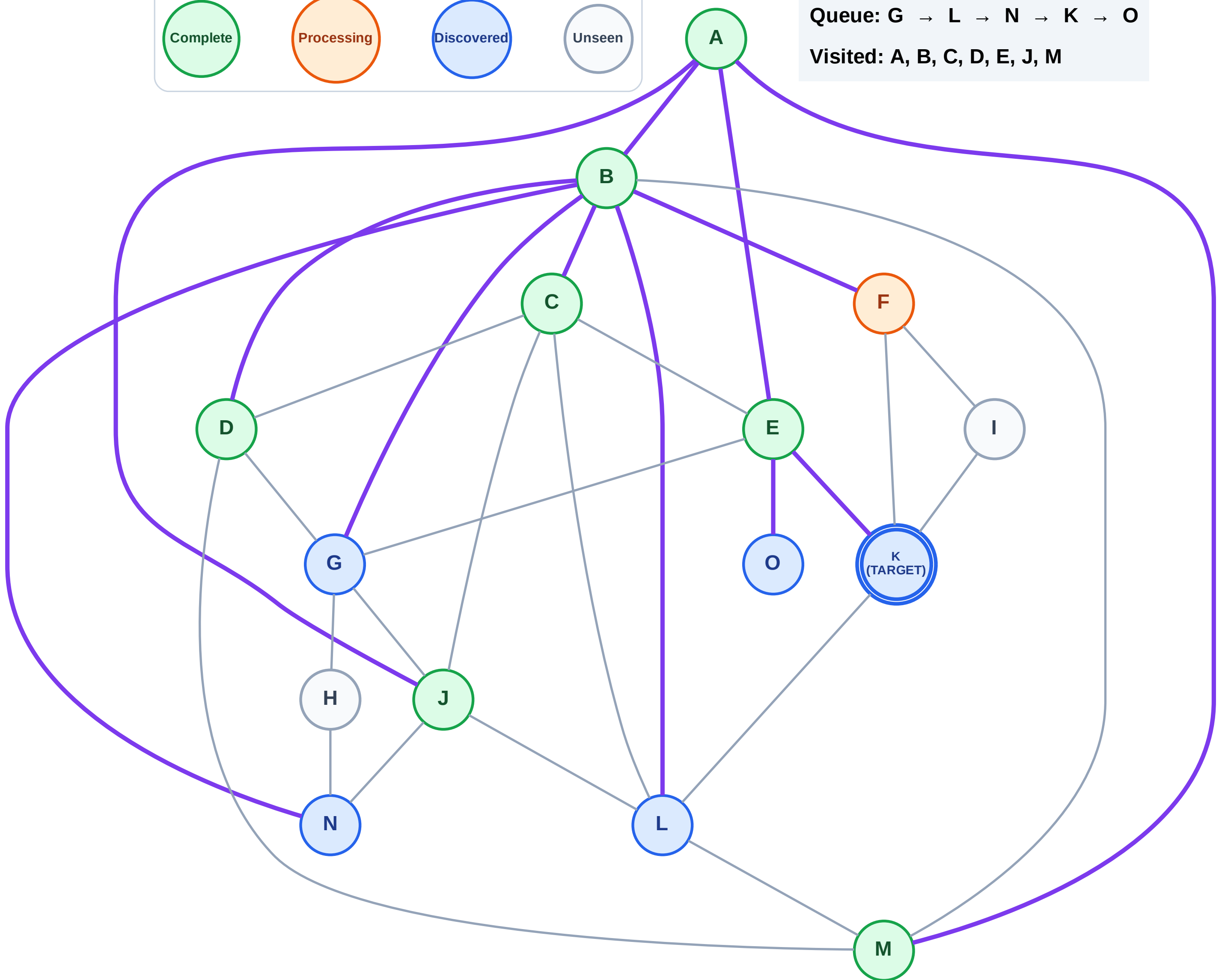
Discovered



Unseen

Queue: G → L → N → K → O

Visited: A, B, C, D, E, J, M



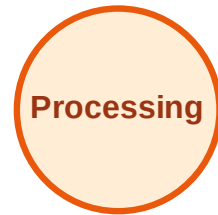
BFS Traversal

Step 17: Discover I from F

Legend



Complete



Processing



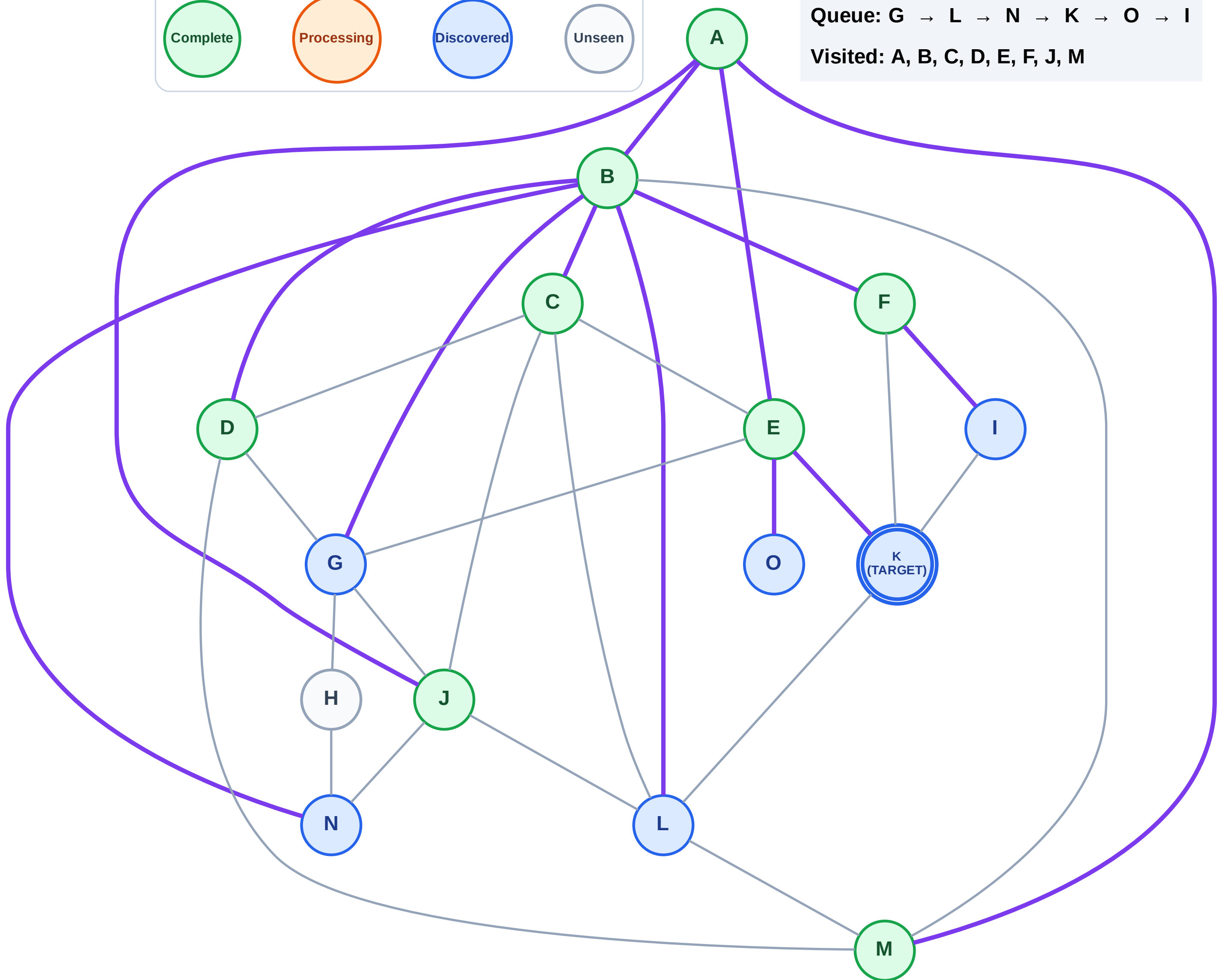
Discovered



Unseen

Queue: G → L → N → K → O → I

Visited: A, B, C, D, E, F, J, M



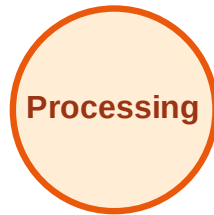
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



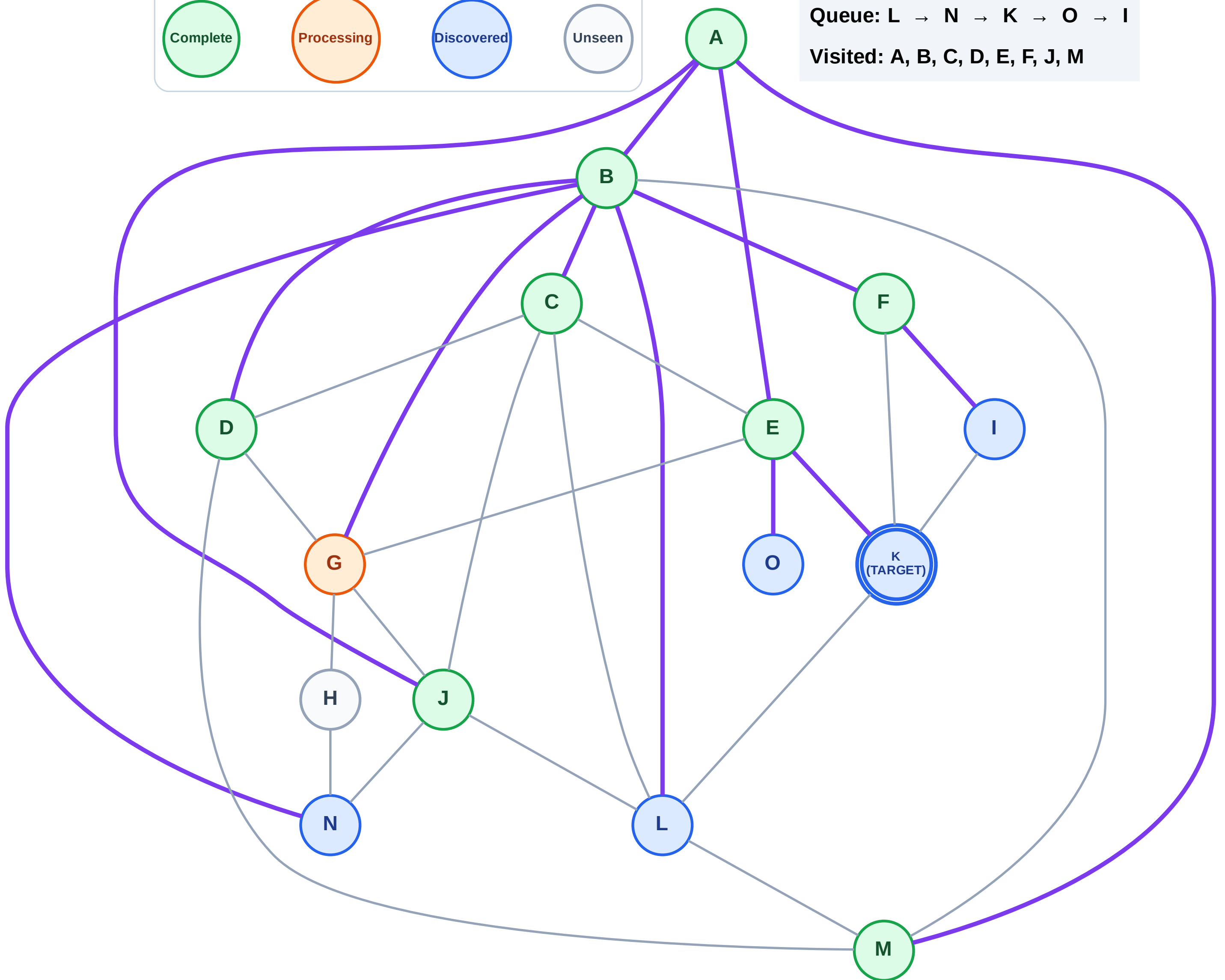
Discovered



Unseen

Queue: L → N → K → O → I

Visited: A, B, C, D, E, F, J, M



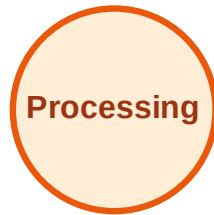
BFS Traversal

Step 19: Discover H from G

Legend



Complete



Processing



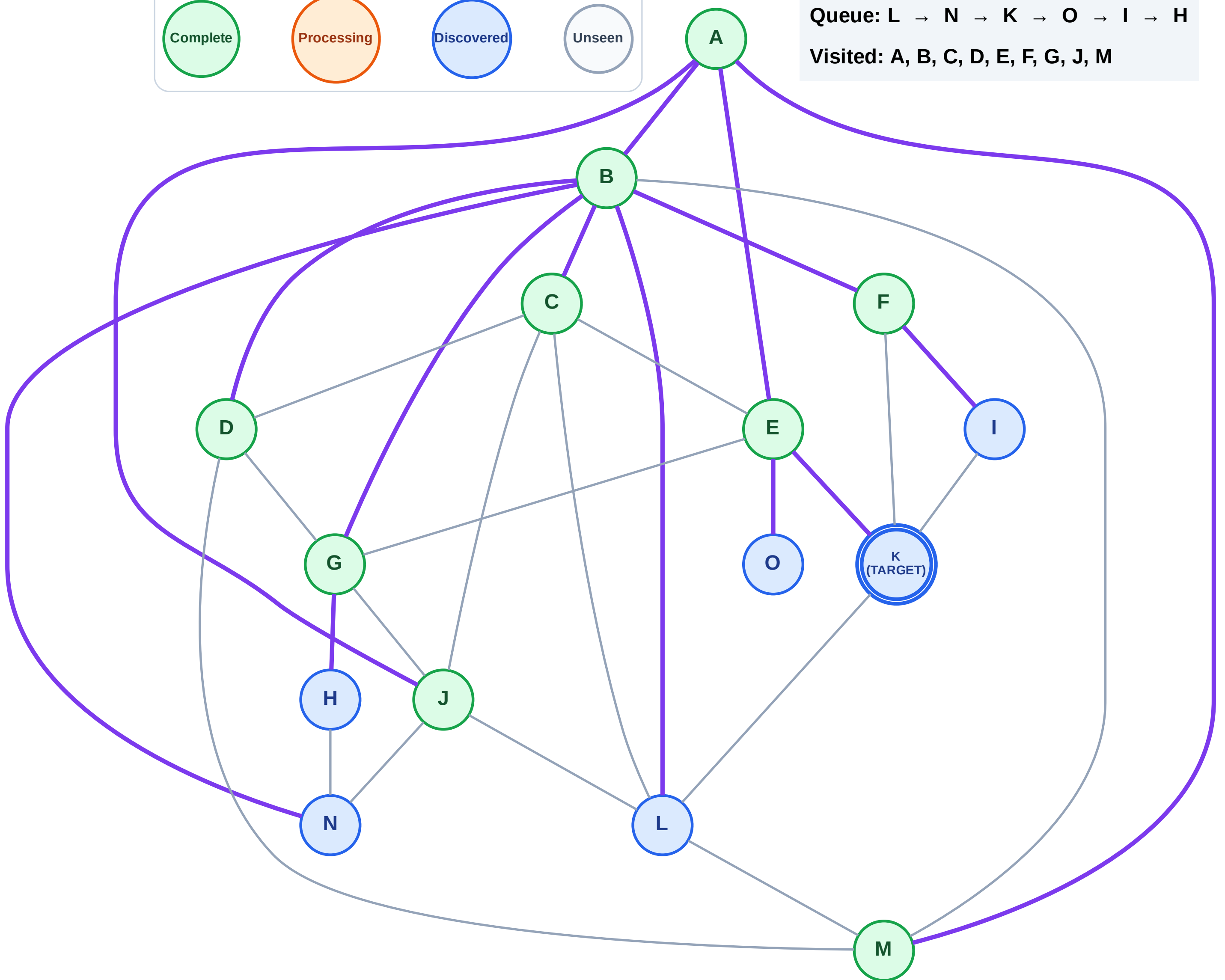
Discovered



Unseen

Queue: L → N → K → O → I → H

Visited: A, B, C, D, E, F, G, J, M



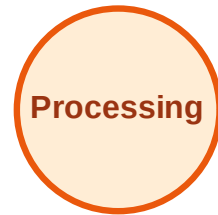
BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



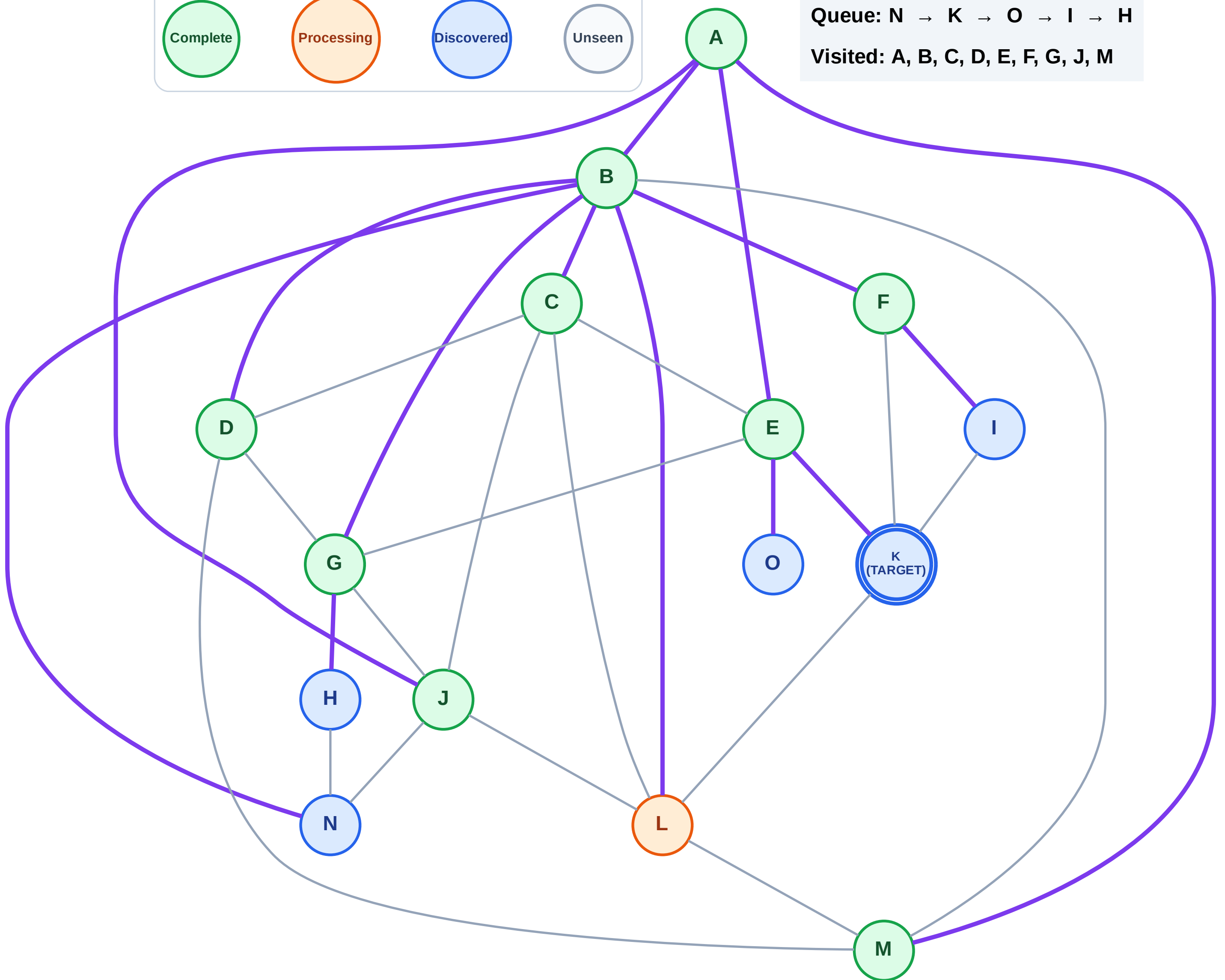
Discovered



Unseen

Queue: N → K → O → I → H

Visited: A, B, C, D, E, F, G, J, M



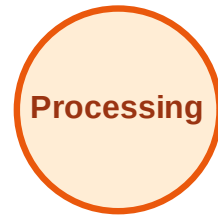
BFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



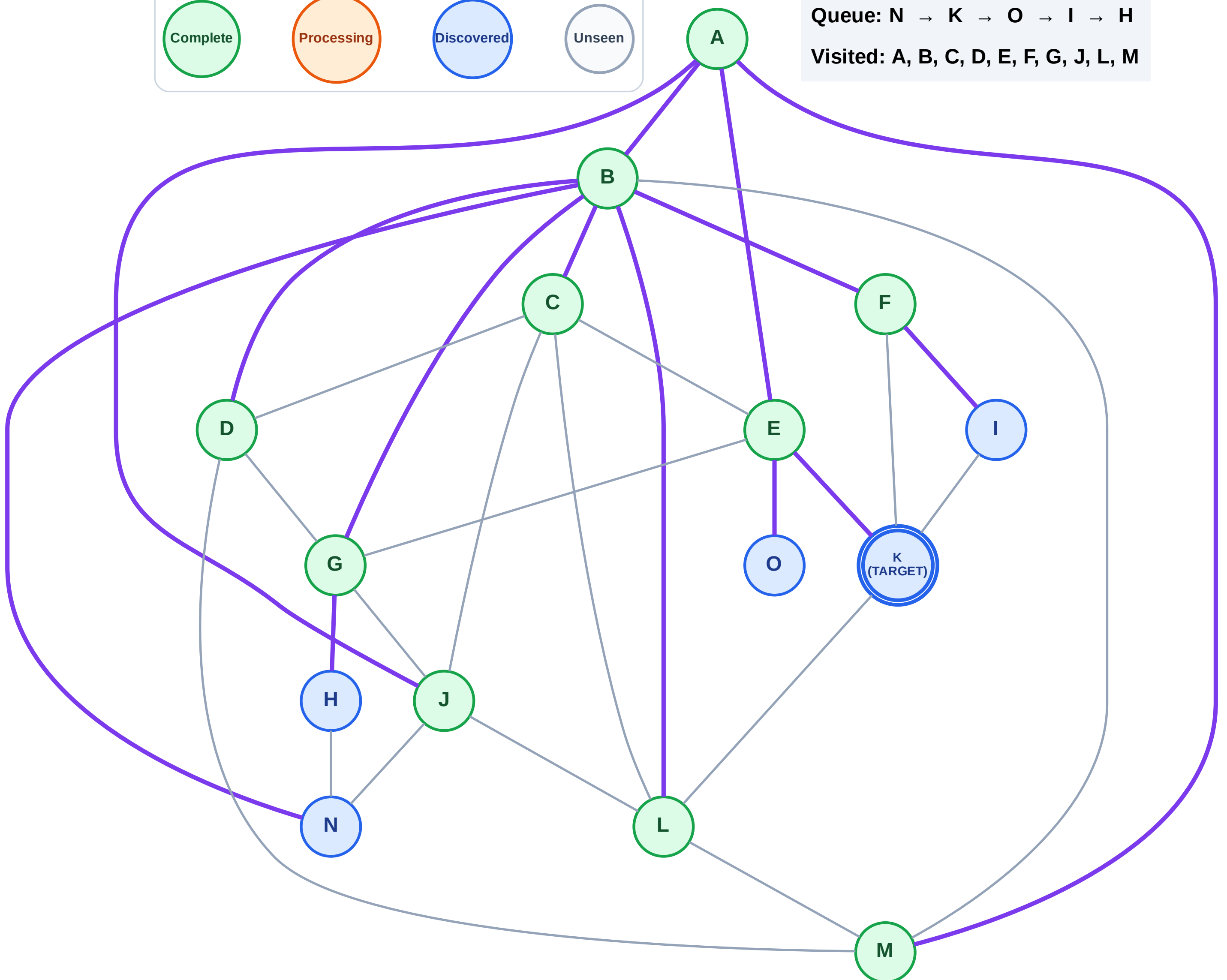
Discovered



Unseen

Queue: N → K → O → I → H

Visited: A, B, C, D, E, F, G, J, L, M



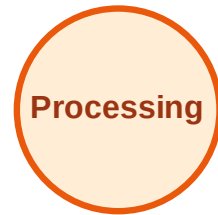
BFS Traversal

Step 22: Process node N

Legend



Complete



Processing



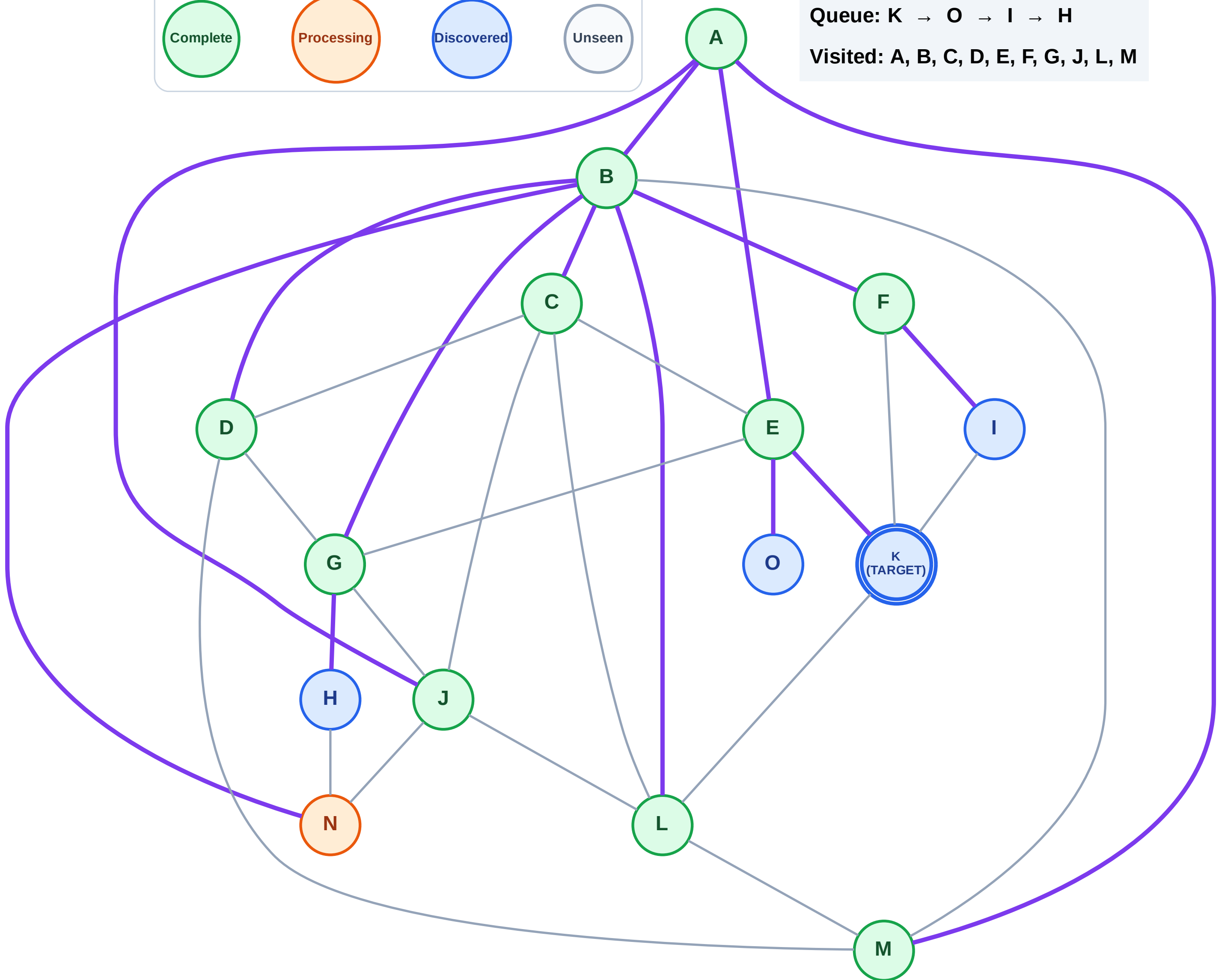
Discovered



Unseen

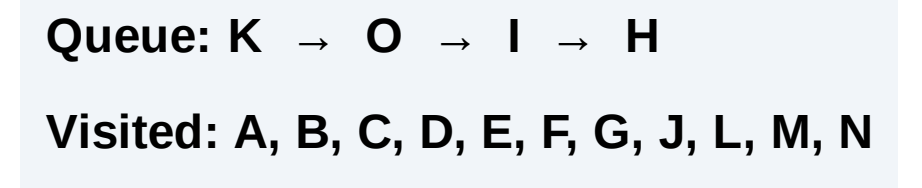
Queue: K → O → I → H

Visited: A, B, C, D, E, F, G, J, L, M



Step 23: No new neighbors from N

Step 23: No new neighbors from N



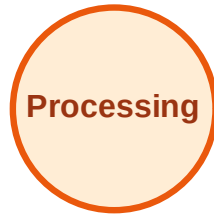
BFS Traversal

Step 24: Process node K

Legend



Complete



Processing



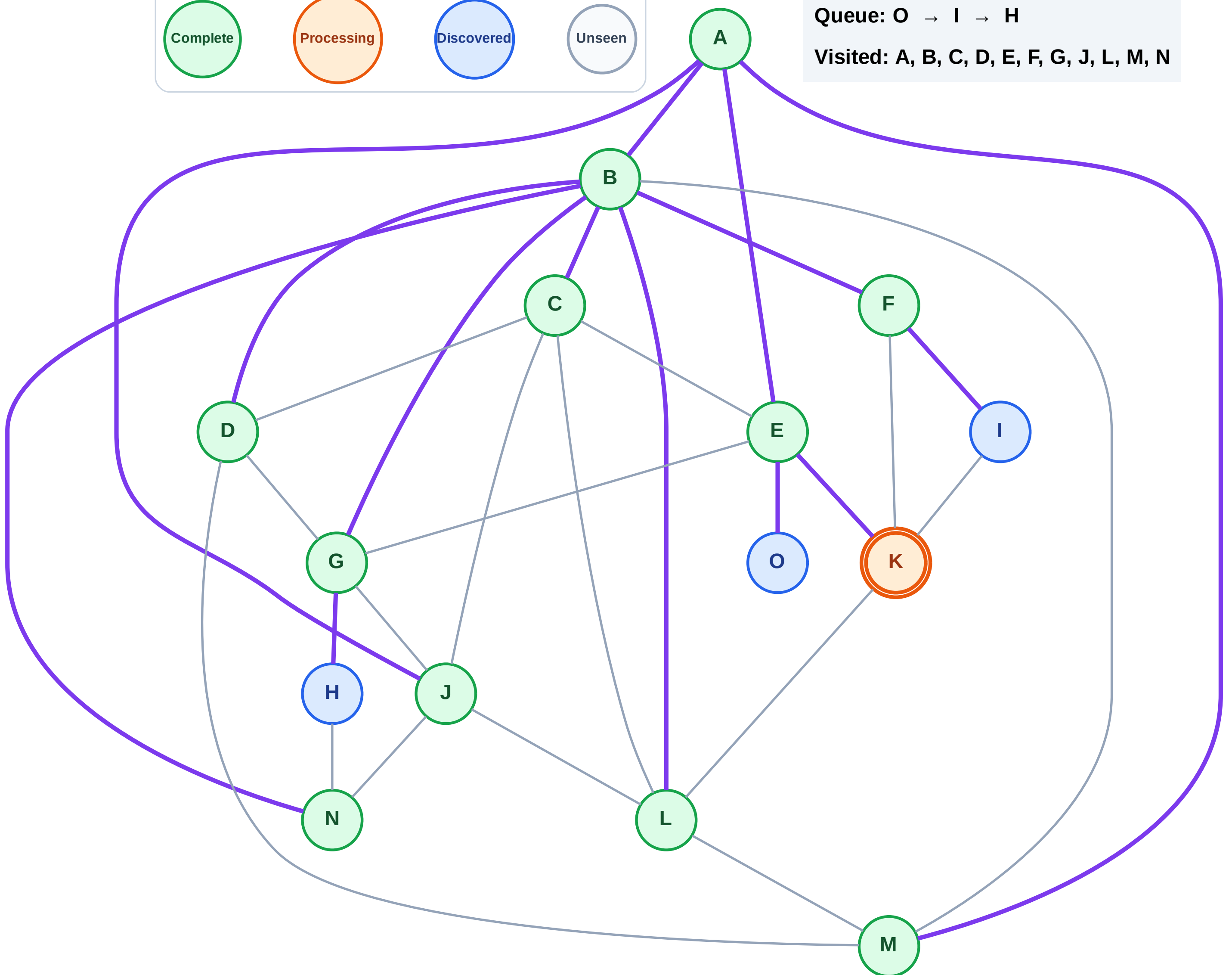
Discovered



Unseen

Queue: O → I → H

Visited: A, B, C, D, E, F, G, J, L, M, N



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

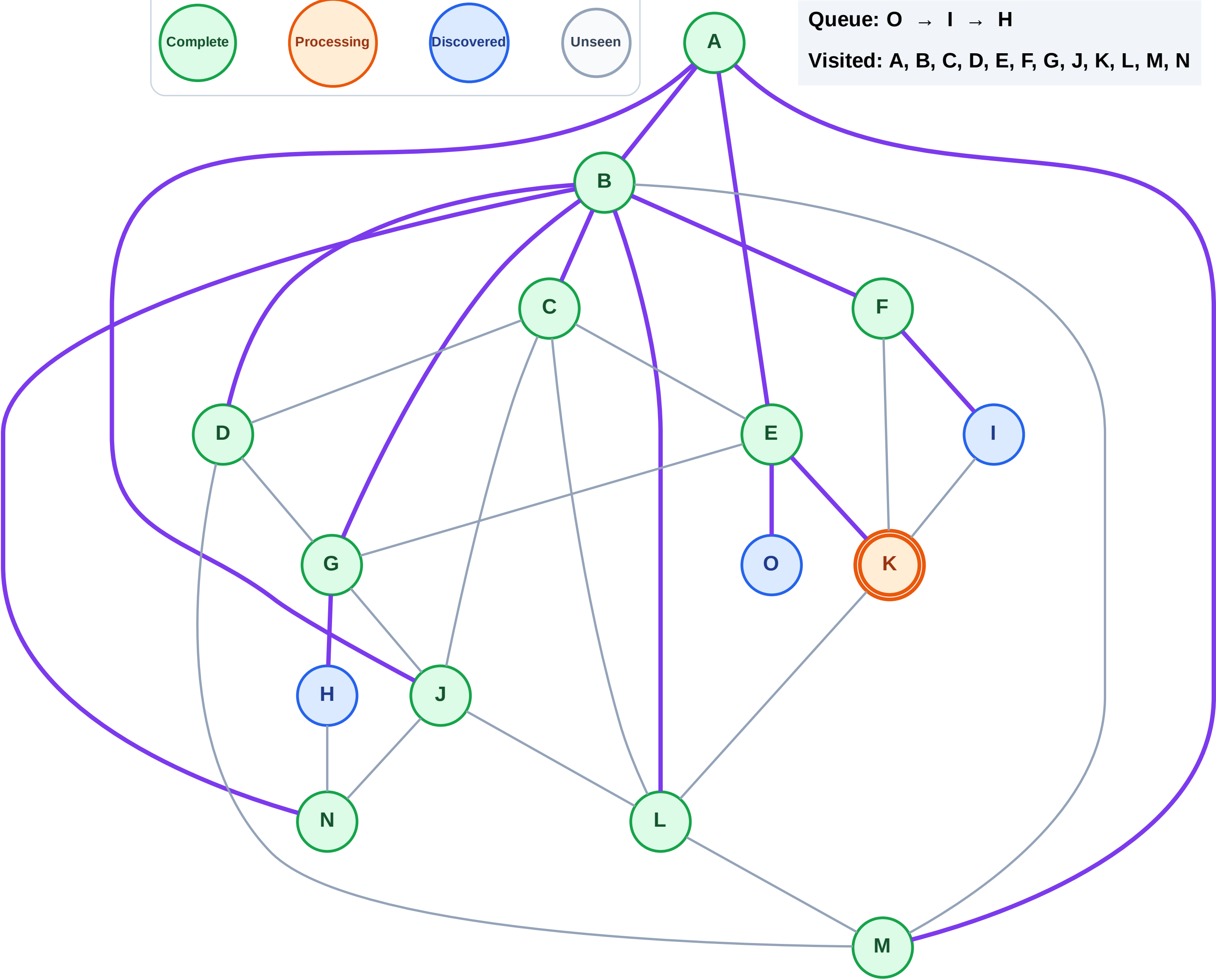
Processing

Discovered

Unseen

Queue: O → I → H

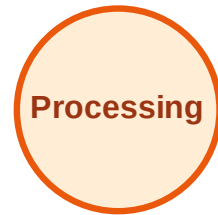
Visited: A, B, C, D, E, F, G, J, K, L, M, N



DFS Traversal

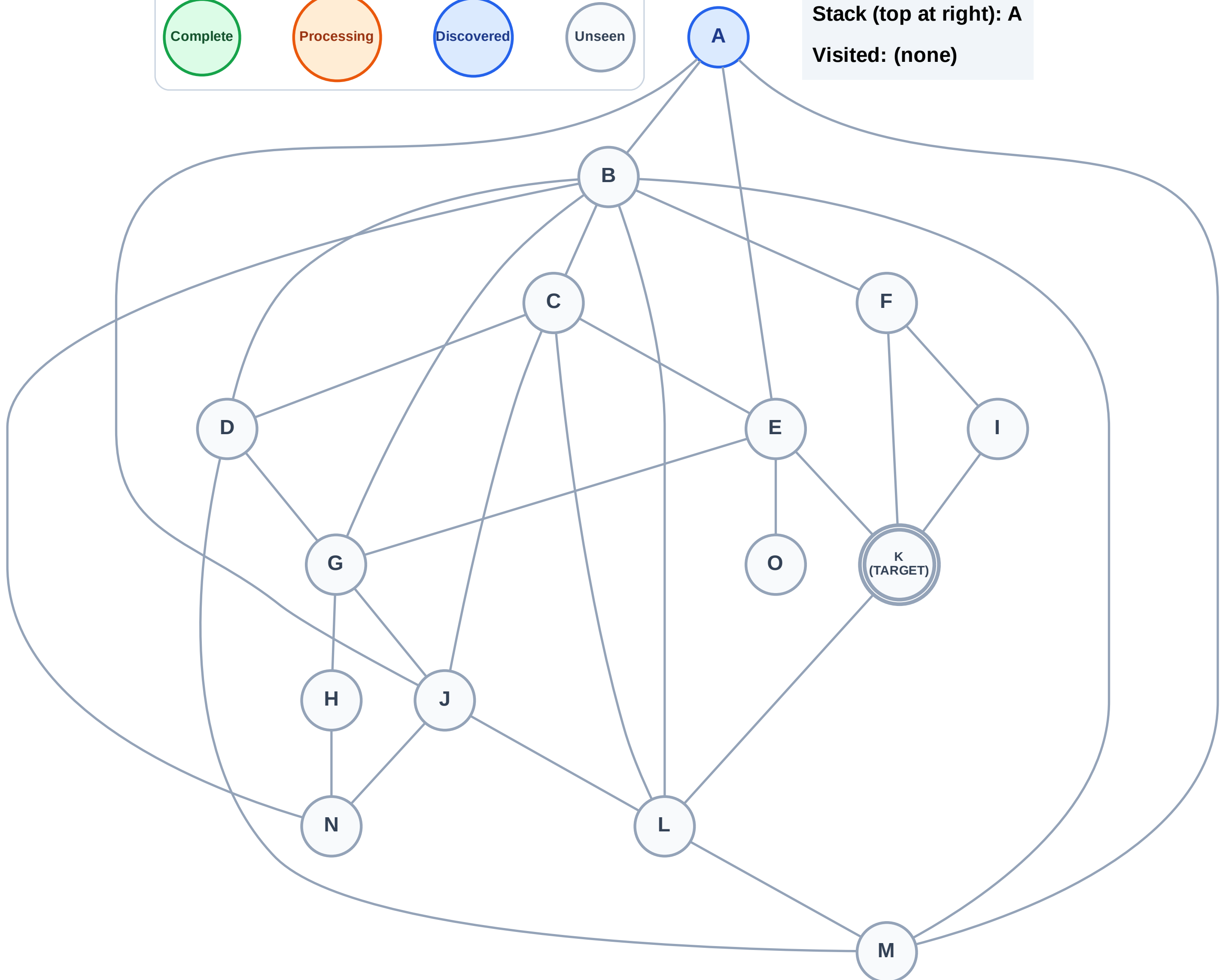
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

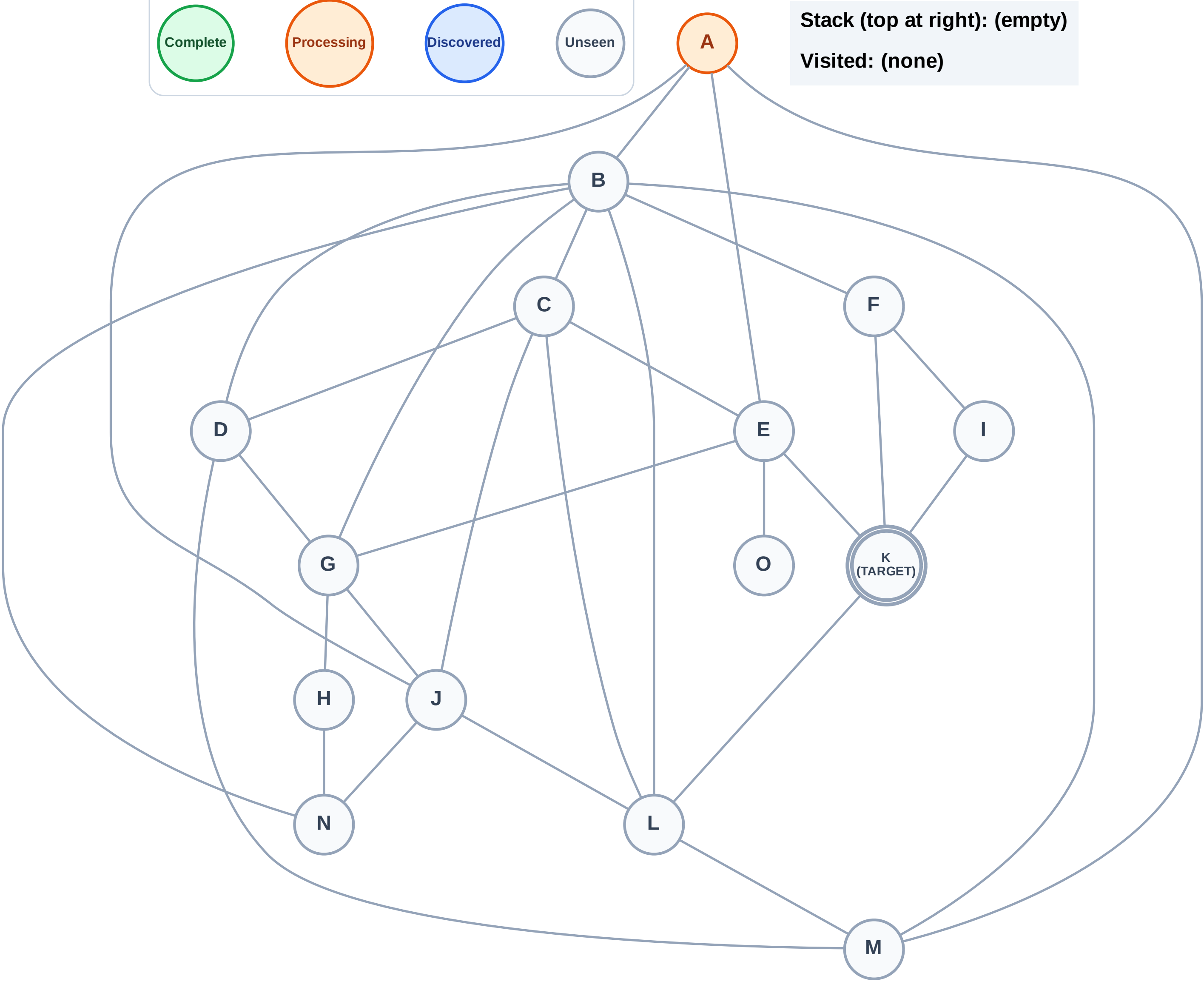
Complete

Processing

Discovered

Unseen

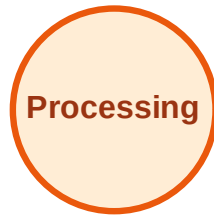
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

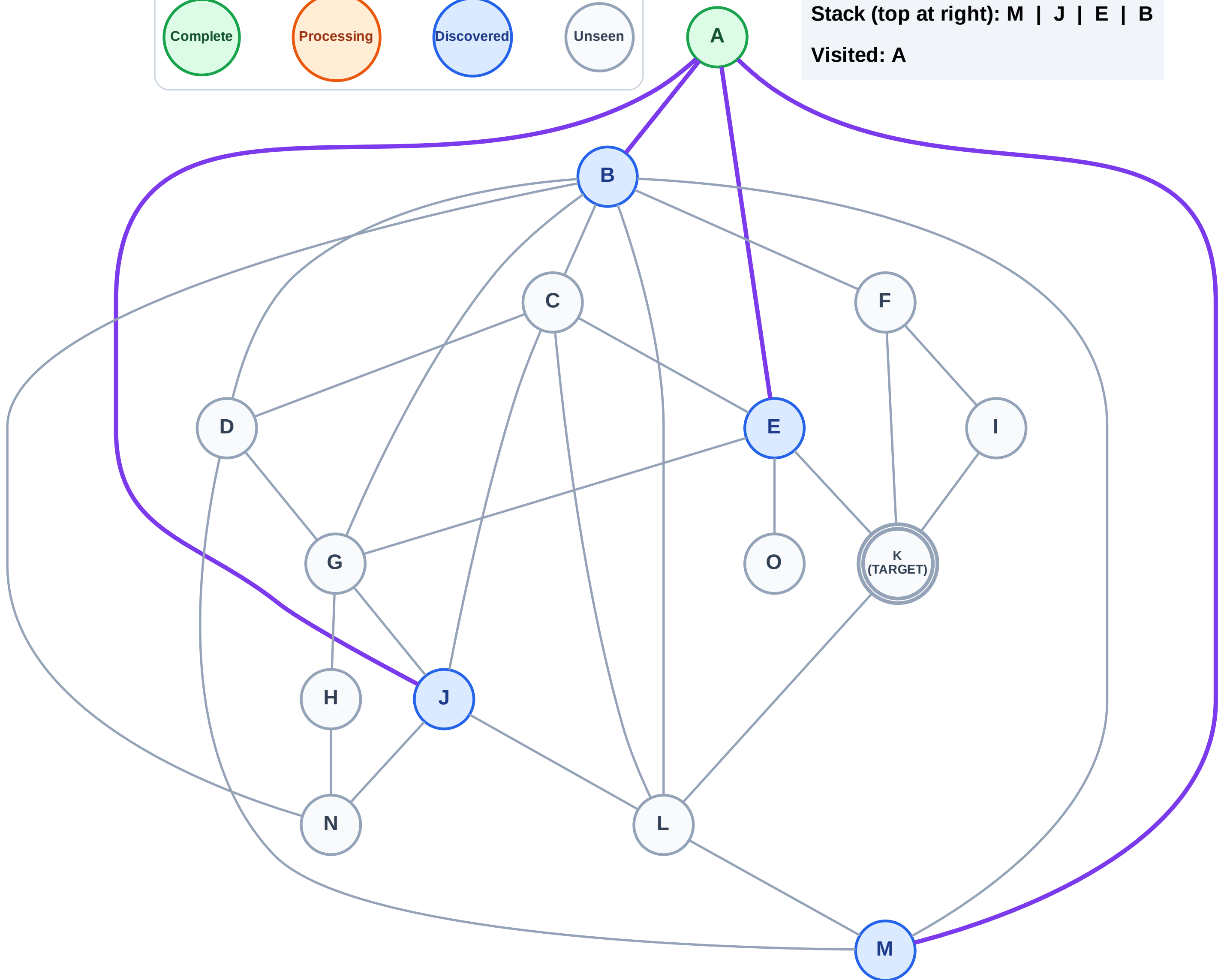
Step 3: Discover M, J, E, B from A

Legend



Stack (top at right): M | J | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

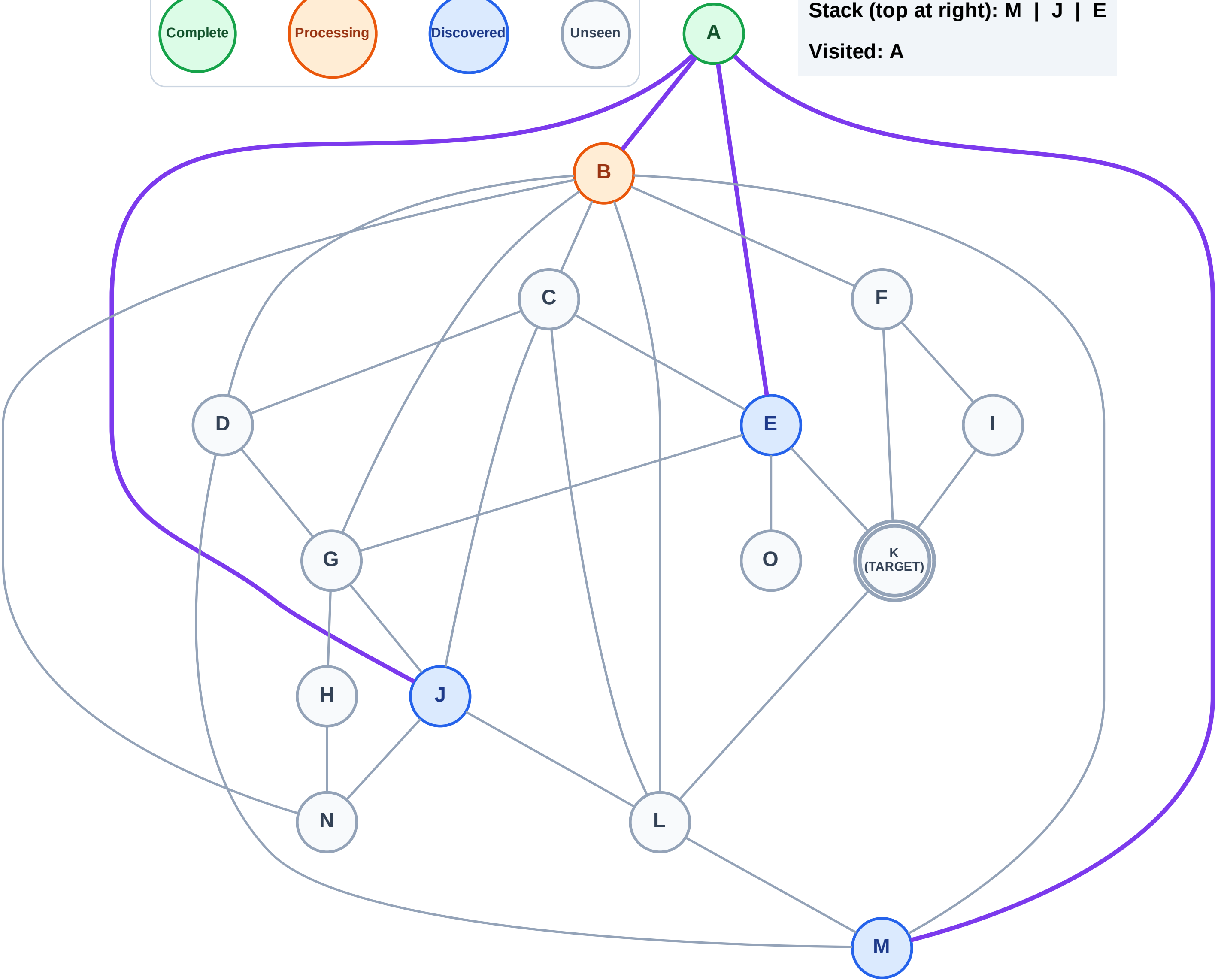
Complete

Processing

Discovered

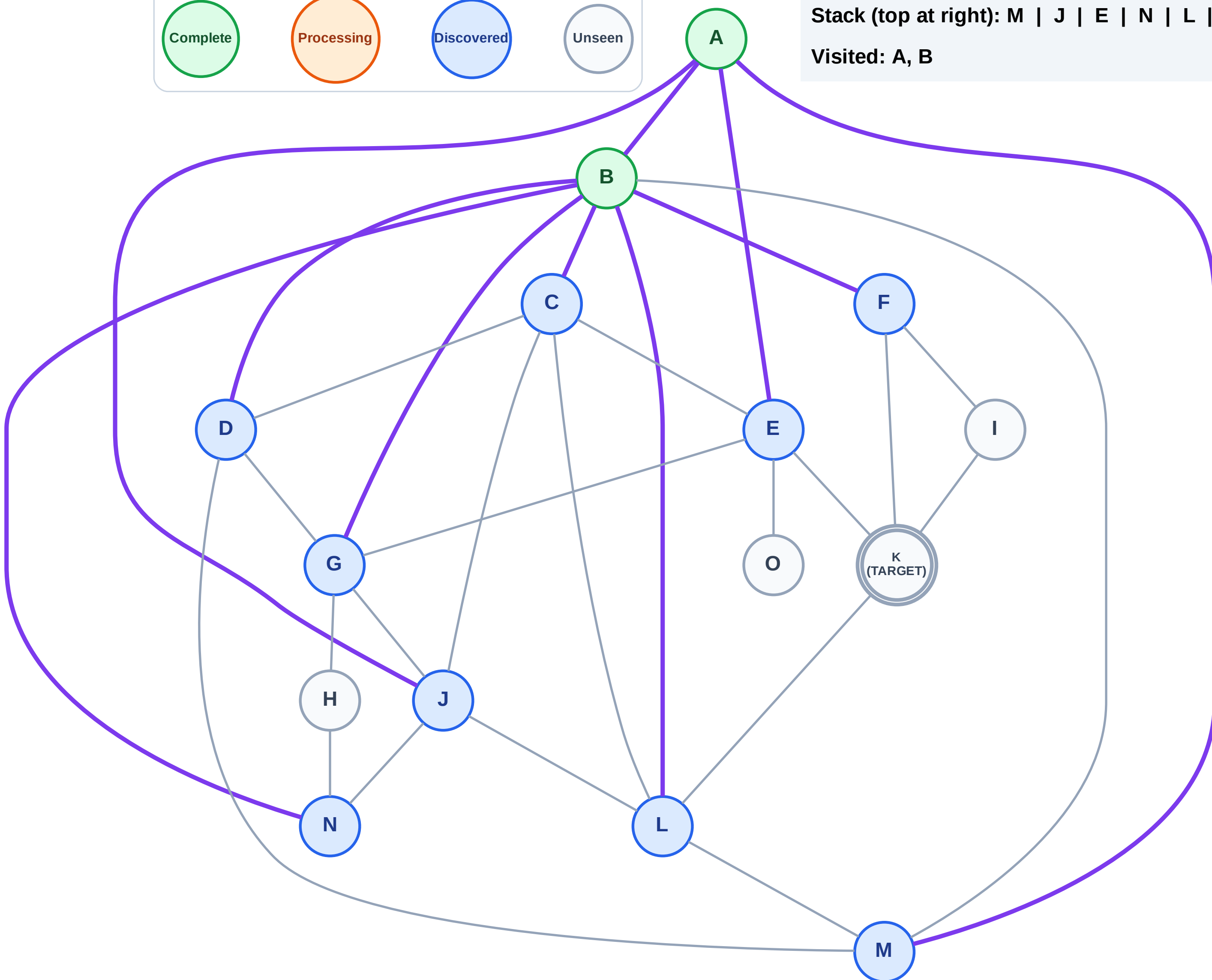
Unseen

Stack (top at right): M | J | E
Visited: A



Step 5: Discover N, L, G, F, D, C from B

Step 5: Discover N, L, G, F, D, C from B



DFS Traversal

Step 6: Process node C

Legend

Complete

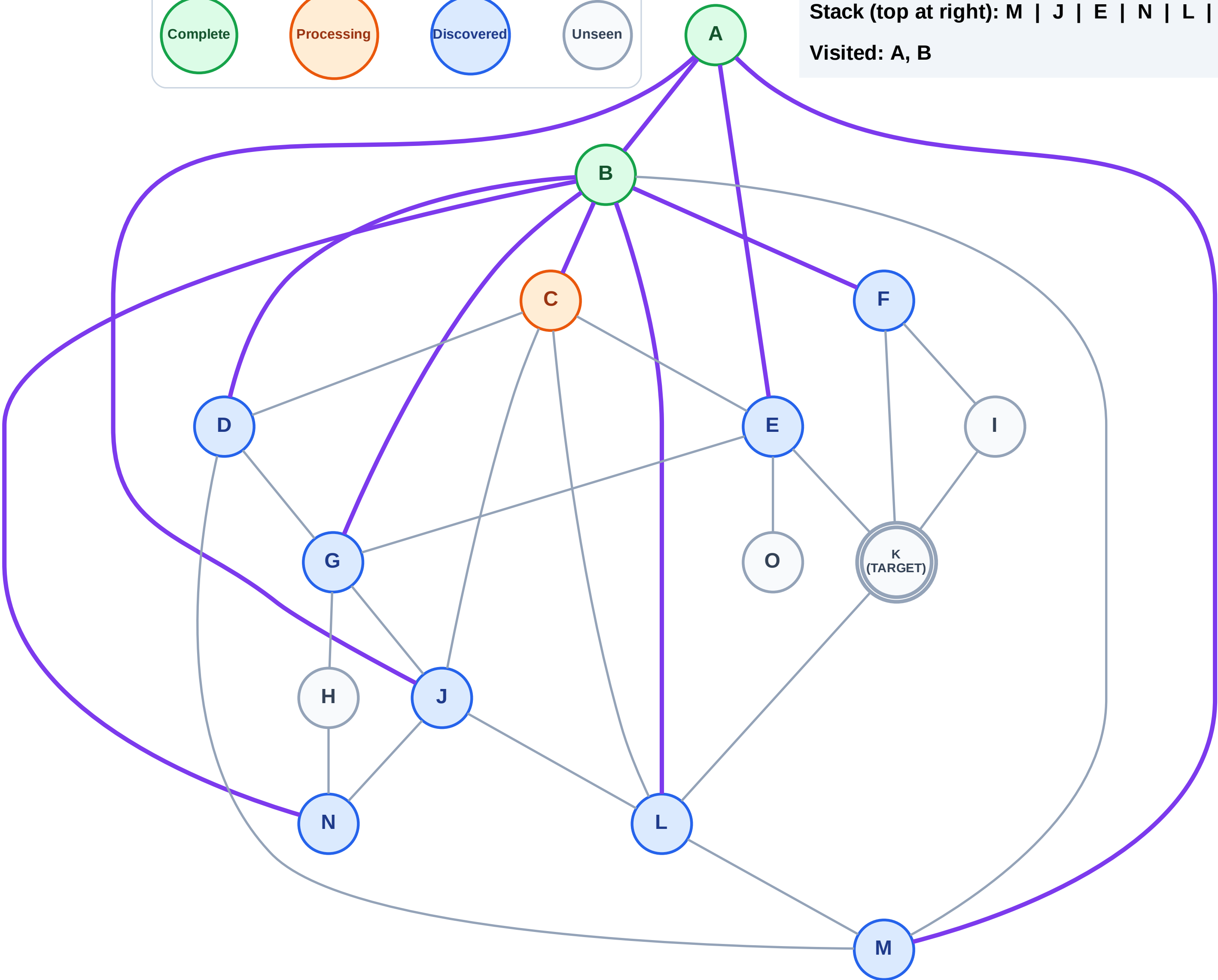
Processing

Discovered

Unseen

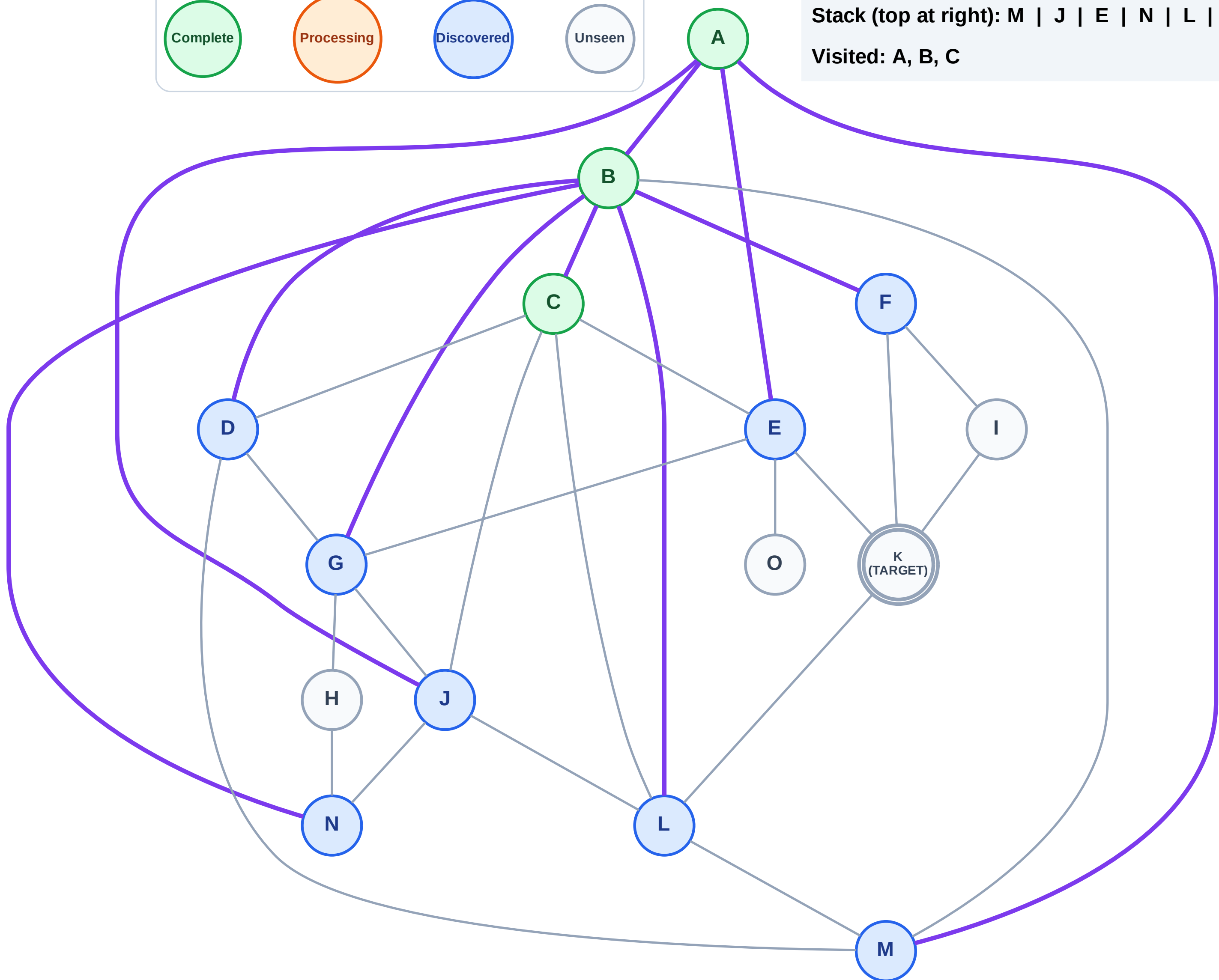
Stack (top at right): M | J | E | N | L | G | F | D

Visited: A, B



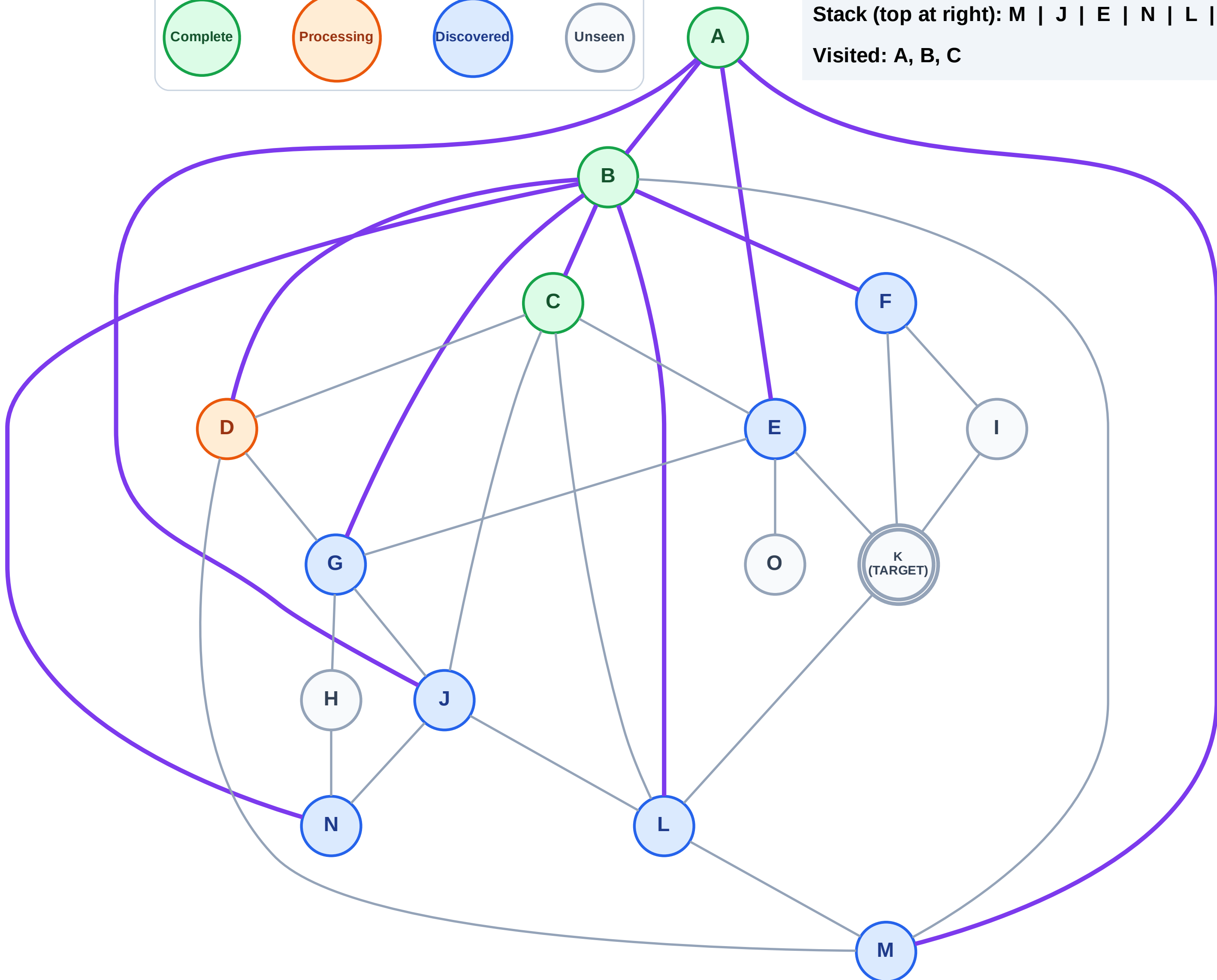
Step 7: No new neighbors from C

Step 7: No new neighbors from C



Step 8: Process node D

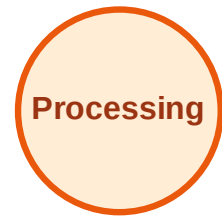
Step 8: Process node D



DFS Traversal

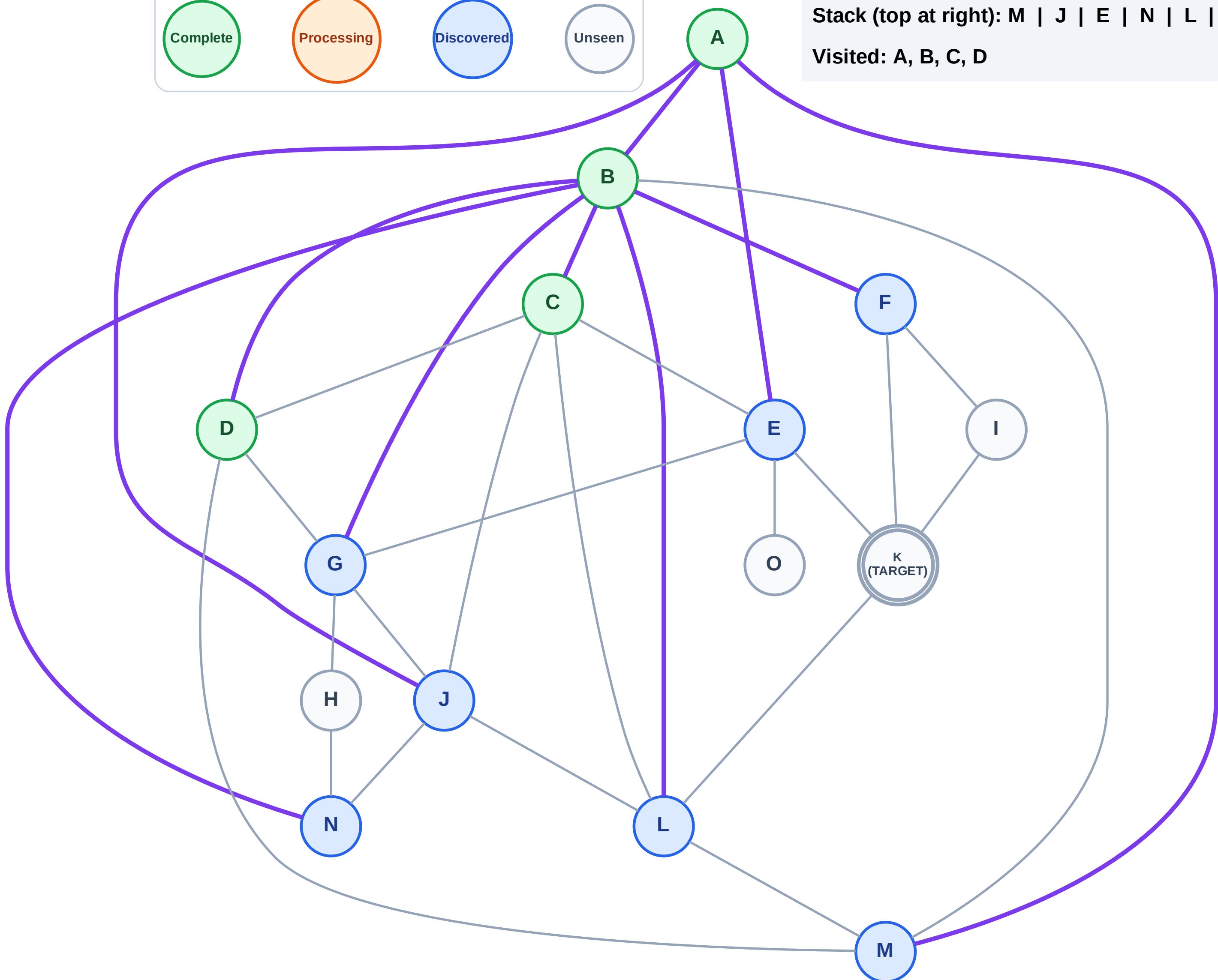
Step 9: No new neighbors from D

Legend



Stack (top at right): M | J | E | N | L | G | F

Visited: A, B, C, D



DFS Traversal

Step 10: Process node F

Legend

Complete

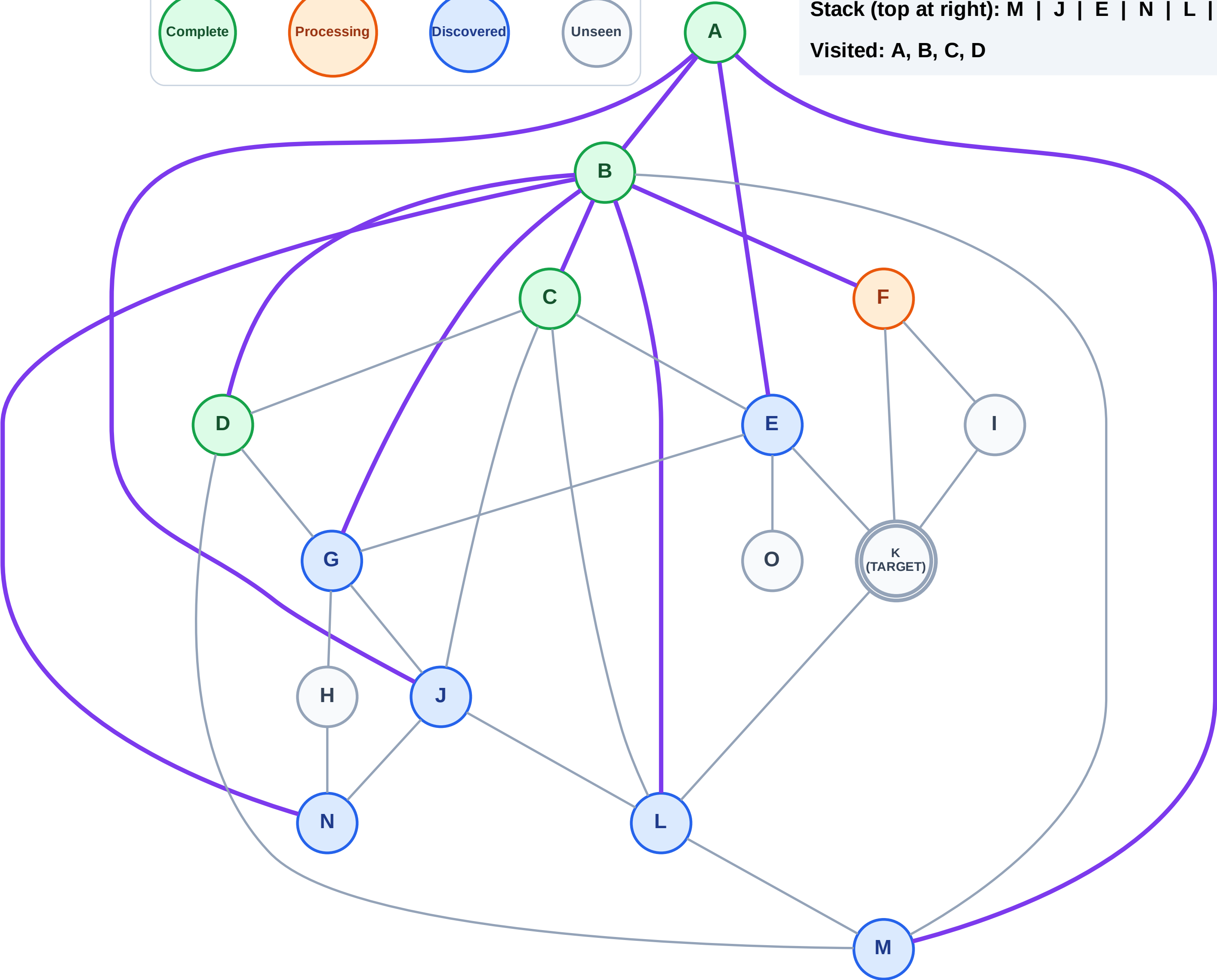
Processing

Discovered

Unseen

Stack (top at right): M | J | E | N | L | G

Visited: A, B, C, D



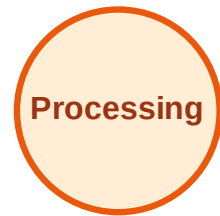
DFS Traversal

Step 11: Discover K, I from F

Legend



Complete



Processing



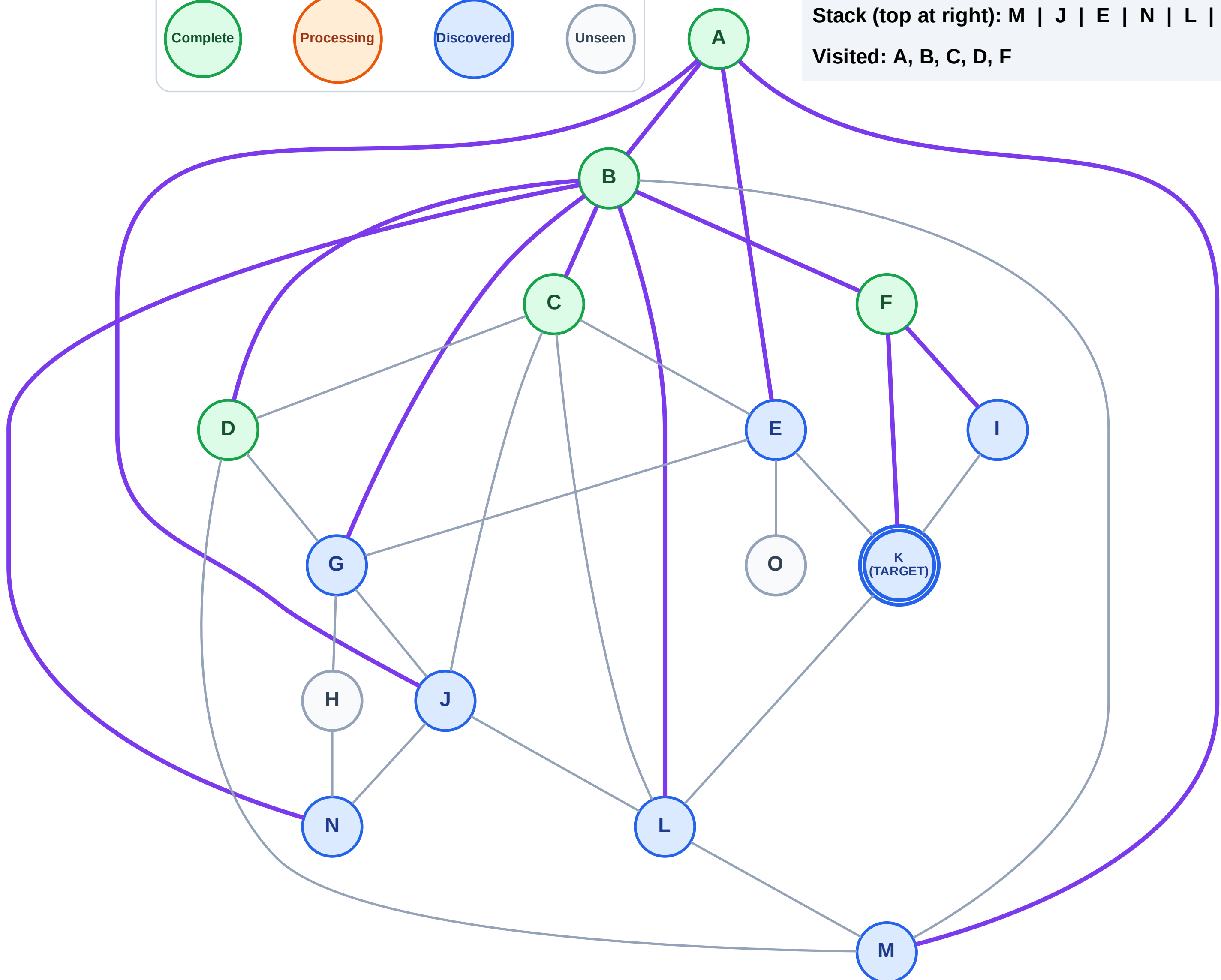
Discovered



Unseen

Stack (top at right): M | J | E | N | L | G | K | I

Visited: A, B, C, D, F



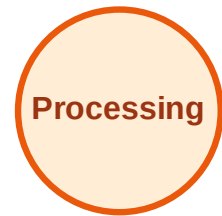
DFS Traversal

Step 12: Process node I

Legend



Complete



Processing



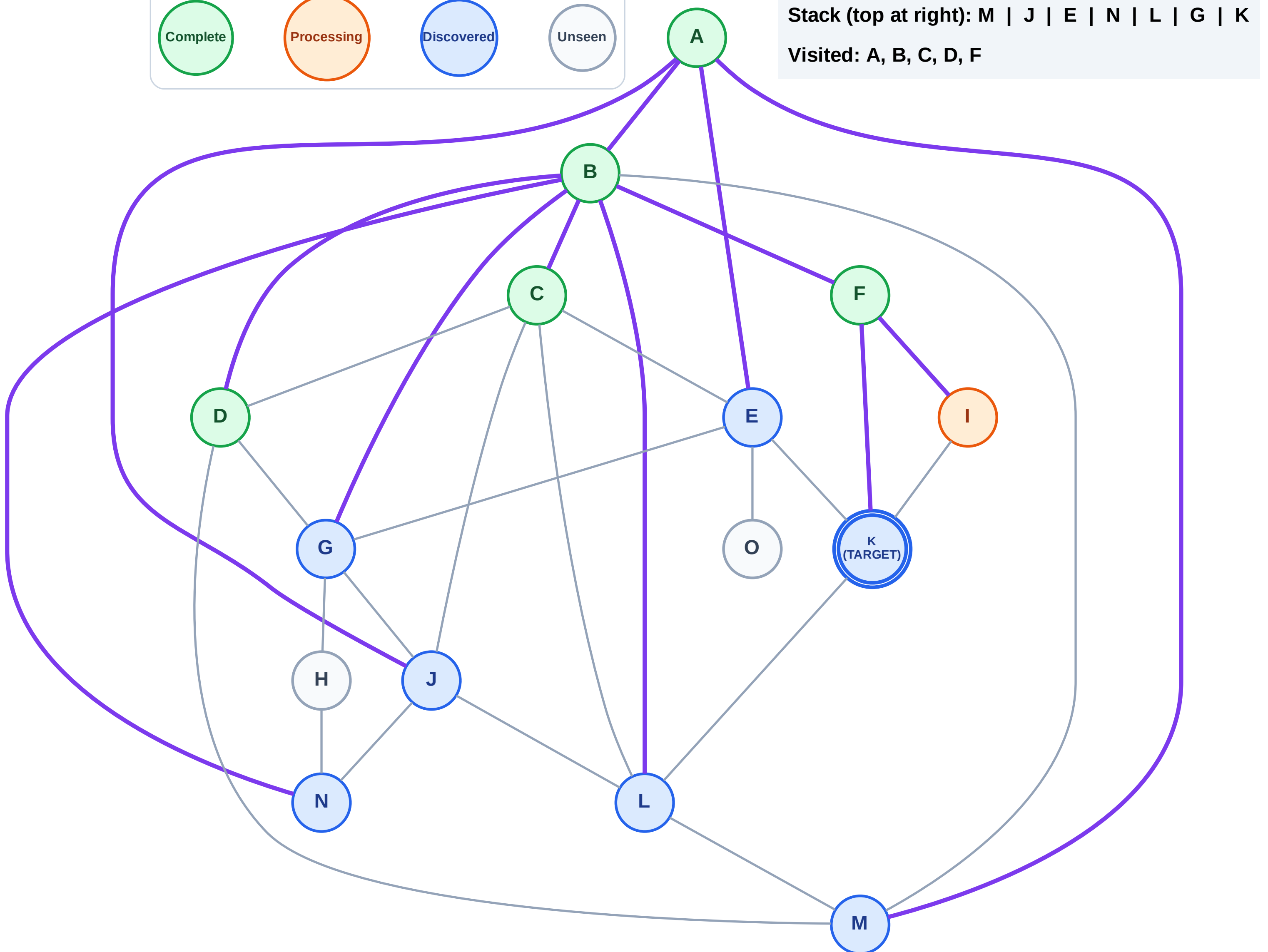
Discovered



Unseen

Stack (top at right): M | J | E | N | L | G | K

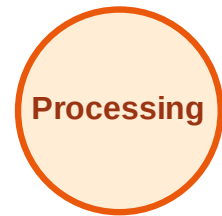
Visited: A, B, C, D, F



DFS Traversal

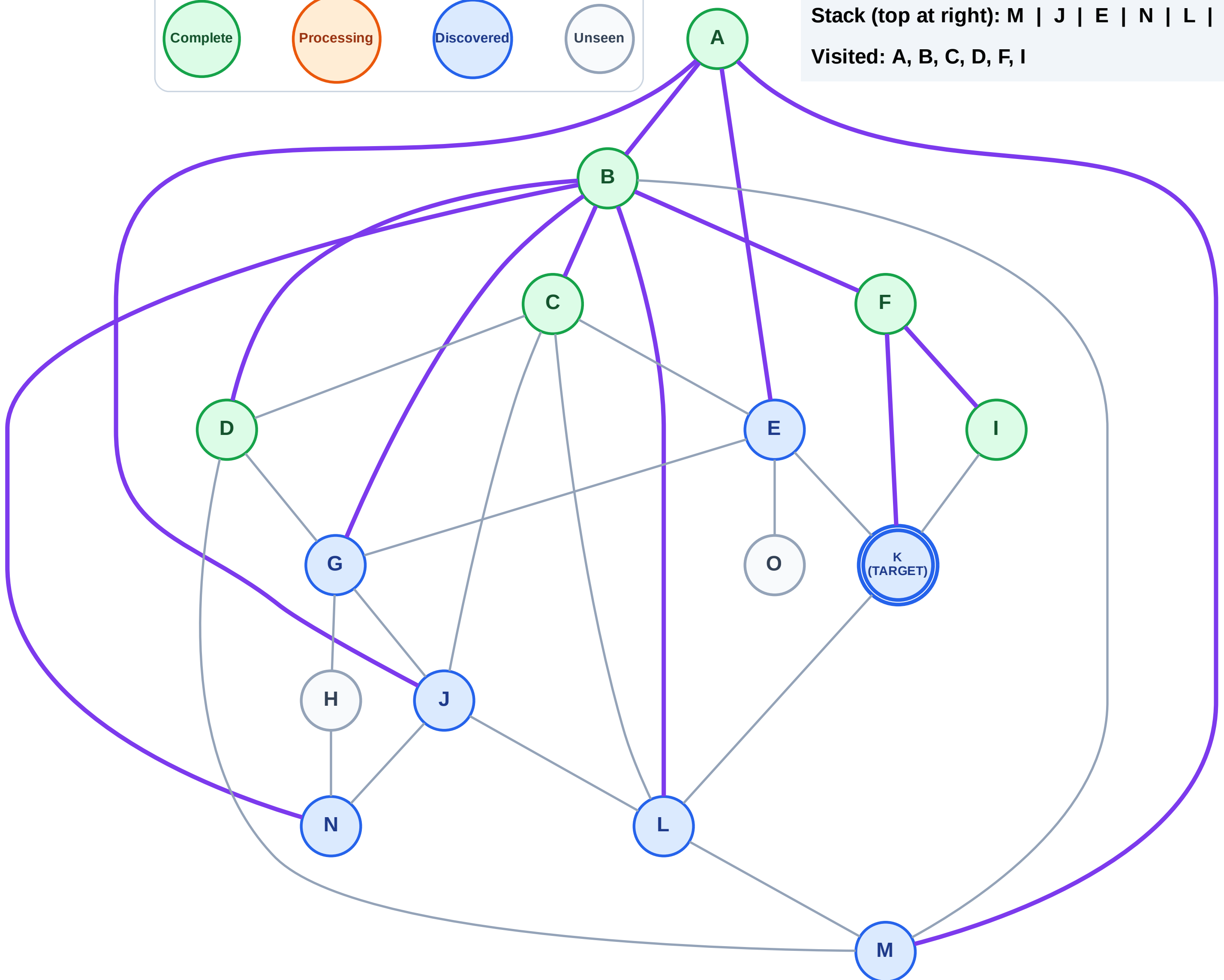
Step 13: No new neighbors from I

Legend



Stack (top at right): M | J | E | N | L | G | K

Visited: A, B, C, D, F, I



DFS Traversal

Step 14: Process node K

Legend

Complete

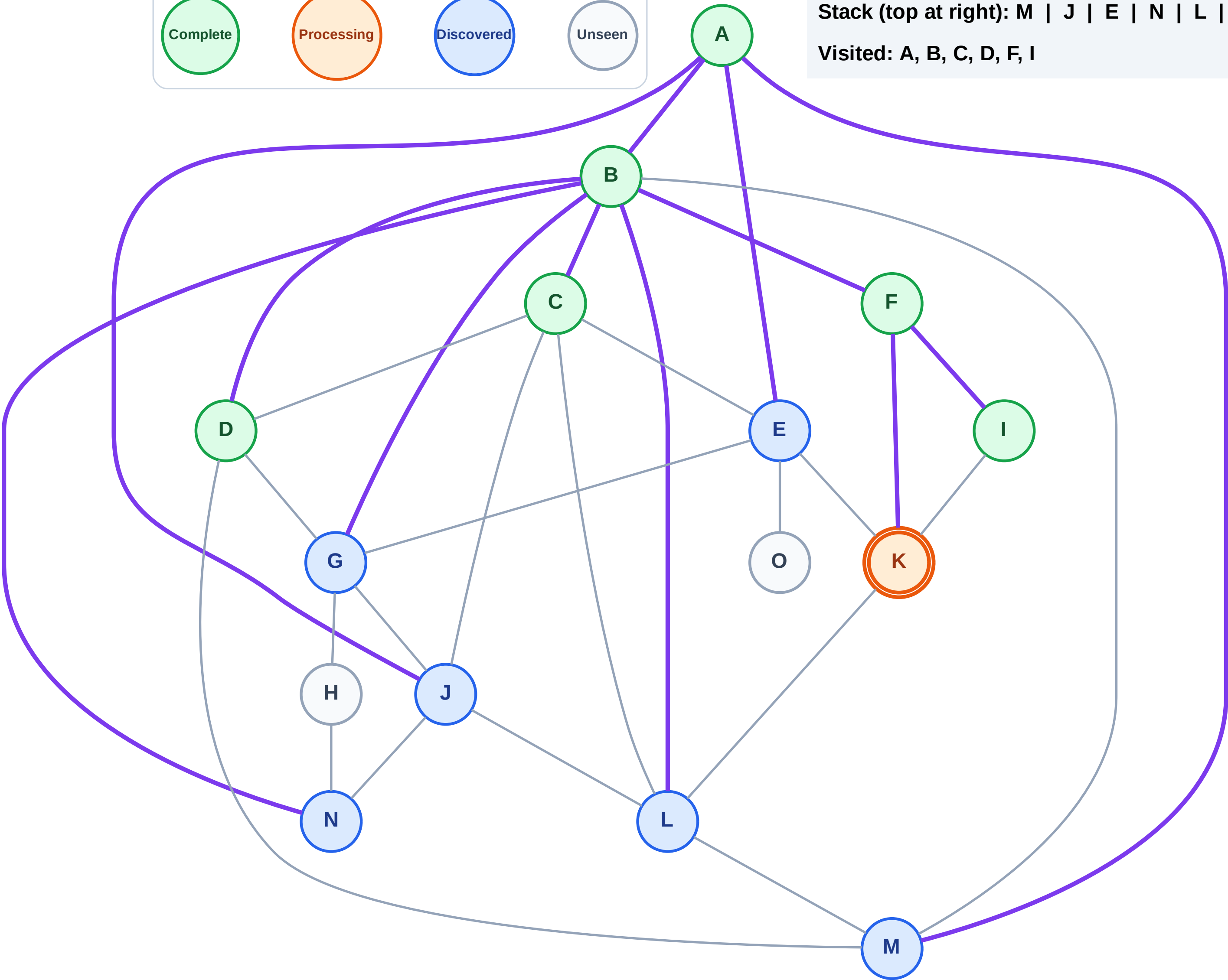
Processing

Discovered

Unseen

Stack (top at right): M | J | E | N | L | G

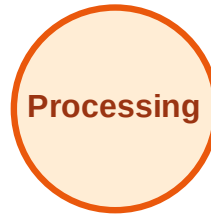
Visited: A, B, C, D, F, I



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend



Stack (top at right): M | J | E | N | L | G

Visited: A, B, C, D, F, I, K

